Supplementary Material
Enforcement of Mass Conservation and Non-Negativity in Statistical Surrogates of the Activated Sludge Process

1 Scope and Reporting Convention

This supplement reports detailed supporting results for the study. The benchmark contains 13 statistical surrogates, eleven nested in-distribution sample totals, five persistent outer folds, and two separately simulated out-of-distribution regimes. Raw and projected predictions always refer to the same observations. No out-of-distribution response entered fitting, preprocessing, hyperparameter selection, or normalization.

For component j, the reported component nRMSE is

$$\mathrm{nRMSE}_j = \frac{\sqrt{n^{-1}\sum_{i=1}^{n}(\hat{c}_{ij} - c_{ij})^2}}{\sigma_j^{(\mathcal{T})}}, \tag{1}$$

where $\sigma_j^{(\mathcal{T})}$ is the population SD of that component in the applicable training reference. For a derived composite $q \in \{\mathrm{COD}, \mathrm{TN}, \mathrm{TP}, \mathrm{TSS}\}$, the requested composite-specific counterpart is

$$\mathrm{nRMSE}_q = \frac{\sqrt{n^{-1}\sum_{i=1}^{n}(\hat{y}_{iq} - y_{iq})^2}}{\sigma_q^{(\mathcal{T})}}. \tag{2}$$

The in-distribution denominator is estimated on the applicable outer-training partition; the out-of-distribution denominator is estimated once on all 10,000 in-distribution targets. Composite nRMSE is a secondary, composite-specific normalization and is not the primary nRMSE obtained by averaging squared standardized errors across all 20 effluent components.

In-distribution values are the arithmetic mean $\pm$ sample SD across five outer folds. Each Δ is calculated within a paired raw/projected fold before those five differences are summarized. Out-of-distribution values are single descriptive summaries over 600 cases and therefore have no fold SD. No hypothesis tests, confidence intervals, or inferential rank claims are reported.

2 Benchmark Contract and Dataset Verification

The 22 inputs were hydraulic retention time, aeration, and 20 influent concentrations. Each surrogate jointly or componentwise predicted the same 20 effluent concentrations in the fixed order S_O, S_F, S_A, S_{NH4}, S_{NO2}, S_{NO3}, S_{N2}, S_{PO4}, S_I, S_{ALK}, X_I, X_S, X_H, X_{PAO}, X_{PP}, X_{PHA}, X_{AOB}, X_{NOB}, X_{MeP}, and X_{MeOH}. COD, TN, TP, and TSS were calculated after prediction through the same fixed linear composition map for raw and projected outputs.

The surrogate roster comprised XGBoost, LightGBM, CatBoost, AdaBoost, Random Forest, Extra Trees, SVR, k-NN, PLS, Multi-task Elastic Net, Multi-task Lasso, MLP, and TabNet. PLS and the two multi-task linear surrogates form the prespecified Interpretable surrogates category because they expose one global coefficient structure; this label does not imply causal interpretation.

Table S1: Mechanistic dataset generation and invariant verification. The invariant residual is $\|Ac_{out} - Ac_{in}\|_2$.

Dataset	Attempted	Retained	Acceptance (%)	Maximum residual	Minimum component
In-distribution	10,000	10,000	100.0	4.90×10^{-11}	8.75×10^{-4}
Mild out-of-distribution	600	600	100.0	5.67×10^{-11}	1.62×10^{-3}
Severe out-of-distribution	600	600	100.0	2.91×10^{-10}	1.01×10^{-3}

The in-distribution accuracy summary contains $13 \times 11 \times 5 \times 2 = 1{,}430$ surrogate–size–fold–prediction rows. Component and composite summaries contain the corresponding 20-component and four-composite expansions. Tuning completed five 100-trial outer-fold studies and one separate 100-trial full-data study per surrogate, for

7,800 completed trials. The out-of-distribution benchmark used one full-data refit per surrogate and retained 600 mechanistic cases per regime.

3 Detailed In-Distribution Predictive Performance

Table S2 gives the same standardized error scale as the primary main-text metric at the individual-component level. The table contains all 260 surrogate–component combinations at $N = 10{,}000$ and preserves raw/projected pairing. Projection lowered mean component nRMSE in 188 combinations (72.3%) and for a majority of components in 12 of 13 surrogates; AdaBoost was the exception at 9 of 20.

Table S2: Full-size in-distribution accuracy by effluent component and surrogate. Values are five-fold mean ± sample SD; Δ is projected minus raw on paired folds.

Component	Unit	Surrogate	Raw nRMSE$_j$	Projected nRMSE$_j$	ΔnRMSE$_j$	Raw R^2_j	Projected R^2_j
S_O	g O_2 m^{-3}	XGBoost	0.124 ± 0.008	0.124 ± 0.008	-0.000 ± 0.000	0.985 ± 0.002	0.985 ± 0.002
S_O	g O_2 m^{-3}	LightGBM	0.114 ± 0.008	0.114 ± 0.008	-0.000 ± 0.000	0.987 ± 0.002	0.987 ± 0.002
S_O	g O_2 m^{-3}	CatBoost	0.101 ± 0.003	0.101 ± 0.003	-0.000 ± 0.000	0.990 ± 0.001	0.990 ± 0.001
S_O	g O_2 m^{-3}	AdaBoost	0.428 ± 0.005	0.429 ± 0.005	$+0.001 \pm 0.001$	0.816 ± 0.007	0.815 ± 0.008
S_O	g O_2 m^{-3}	Random Forest	0.738 ± 0.024	0.740 ± 0.024	$+0.002 \pm 0.001$	0.455 ± 0.012	0.453 ± 0.012
S_O	g O_2 m^{-3}	Extra Trees	0.341 ± 0.007	0.345 ± 0.006	$+0.004 \pm 0.002$	0.884 ± 0.006	0.881 ± 0.007
S_O	g O_2 m^{-3}	SVR	0.207 ± 0.003	0.202 ± 0.003	-0.004 ± 0.001	0.957 ± 0.002	0.959 ± 0.002
S_O	g O_2 m^{-3}	k-NN	0.552 ± 0.017	0.552 ± 0.017	$+0.001 \pm 0.001$	0.696 ± 0.007	0.694 ± 0.008
S_O	g O_2 m^{-3}	PLS	0.533 ± 0.007	0.490 ± 0.007	-0.043 ± 0.006	0.715 ± 0.014	0.759 ± 0.009
S_O	g O_2 m^{-3}	Multi-task Elastic Net	0.486 ± 0.004	0.427 ± 0.008	-0.059 ± 0.008	0.763 ± 0.011	0.817 ± 0.004
S_O	g O_2 m^{-3}	Multi-task Lasso	0.486 ± 0.004	0.427 ± 0.008	-0.059 ± 0.008	0.763 ± 0.011	0.817 ± 0.004
S_O	g O_2 m^{-3}	MLP	0.091 ± 0.013	0.090 ± 0.013	-0.000 ± 0.000	0.992 ± 0.003	0.992 ± 0.003
S_O	g O_2 m^{-3}	TabNet	0.180 ± 0.011	0.182 ± 0.016	$+0.002 \pm 0.010$	0.967 ± 0.004	0.966 ± 0.007
S_F	g COD m^{-3}	XGBoost	0.501 ± 0.034	0.501 ± 0.034	-0.000 ± 0.000	0.744 ± 0.046	0.744 ± 0.046
S_F	g COD m^{-3}	LightGBM	0.521 ± 0.055	0.521 ± 0.055	-0.000 ± 0.000	0.721 ± 0.077	0.721 ± 0.077
S_F	g COD m^{-3}	CatBoost	0.479 ± 0.045	0.479 ± 0.045	-0.000 ± 0.000	0.768 ± 0.032	0.768 ± 0.032
S_F	g COD m^{-3}	AdaBoost	0.763 ± 0.054	1.147 ± 0.280	$+0.384 \pm 0.280$	0.414 ± 0.039	-0.411 ± 0.659
S_F	g COD m^{-3}	Random Forest	0.887 ± 0.100	1.993 ± 0.150	$+1.107 \pm 0.211$	0.214 ± 0.029	-3.118 ± 1.112
S_F	g COD m^{-3}	Extra Trees	0.762 ± 0.070	1.679 ± 0.184	$+0.917 \pm 0.170$	0.417 ± 0.027	-1.898 ± 0.713
S_F	g COD m^{-3}	SVR	0.853 ± 0.105	0.856 ± 0.104	$+0.002 \pm 0.006$	0.273 ± 0.041	0.268 ± 0.043
S_F	g COD m^{-3}	k-NN	0.924 ± 0.088	1.273 ± 0.222	$+0.349 \pm 0.216$	0.145 ± 0.022	-0.682 ± 0.550
S_F	g COD m^{-3}	PLS	0.980 ± 0.096	1.833 ± 0.259	$+0.853 \pm 0.195$	0.038 ± 0.007	-2.383 ± 0.652
S_F	g COD m^{-3}	Multi-task Elastic Net	0.921 ± 0.097	0.917 ± 0.097	-0.003 ± 0.001	0.152 ± 0.017	0.158 ± 0.018
S_F	g COD m^{-3}	Multi-task Lasso	0.921 ± 0.097	0.917 ± 0.097	-0.003 ± 0.001	0.152 ± 0.017	0.158 ± 0.018
S_F	g COD m^{-3}	MLP	0.453 ± 0.082	0.452 ± 0.082	-0.001 ± 0.000	0.793 ± 0.054	0.794 ± 0.054
S_F	g COD m^{-3}	TabNet	0.579 ± 0.178	0.658 ± 0.185	$+0.078 \pm 0.152$	0.657 ± 0.168	0.535 ± 0.268
S_A	g COD m^{-3}	XGBoost	0.293 ± 0.016	0.293 ± 0.017	-0.000 ± 0.000	0.914 ± 0.005	0.914 ± 0.005
S_A	g COD m^{-3}	LightGBM	0.266 ± 0.019	0.266 ± 0.019	-0.000 ± 0.000	0.929 ± 0.007	0.929 ± 0.008
S_A	g COD m^{-3}	CatBoost	0.237 ± 0.023	0.234 ± 0.023	-0.002 ± 0.000	0.944 ± 0.004	0.945 ± 0.004
S_A	g COD m^{-3}	AdaBoost	0.597 ± 0.031	0.600 ± 0.032	$+0.004 \pm 0.003$	0.641 ± 0.034	0.637 ± 0.033
S_A	g COD m^{-3}	Random Forest	0.592 ± 0.045	0.608 ± 0.046	$+0.016 \pm 0.006$	0.648 ± 0.028	0.628 ± 0.030
S_A	g COD m^{-3}	Extra Trees	0.503 ± 0.031	0.517 ± 0.031	$+0.014 \pm 0.006$	0.746 ± 0.010	0.732 ± 0.015
S_A	g COD m^{-3}	SVR	0.418 ± 0.033	0.400 ± 0.034	-0.018 ± 0.001	0.824 ± 0.014	0.840 ± 0.014
S_A	g COD m^{-3}	k-NN	0.733 ± 0.060	0.736 ± 0.061	$+0.003 \pm 0.003$	0.462 ± 0.017	0.457 ± 0.020
S_A	g COD m^{-3}	PLS	0.825 ± 0.060	0.804 ± 0.062	-0.021 ± 0.007	0.319 ± 0.015	0.353 ± 0.018
S_A	g COD m^{-3}	Multi-task Elastic Net	0.805 ± 0.058	0.765 ± 0.061	-0.040 ± 0.005	0.351 ± 0.016	0.414 ± 0.013
S_A	g COD m^{-3}	Multi-task Lasso	0.805 ± 0.058	0.765 ± 0.061	-0.040 ± 0.005	0.351 ± 0.016	0.414 ± 0.013
S_A	g COD m^{-3}	MLP	0.109 ± 0.009	0.106 ± 0.008	-0.003 ± 0.002	0.988 ± 0.000	0.989 ± 0.000
S_A	g COD m^{-3}	TabNet	0.218 ± 0.042	0.220 ± 0.044	$+0.002 \pm 0.010$	0.951 ± 0.019	0.950 ± 0.019
S_{NH4}	g N m^{-3}	XGBoost	0.124 ± 0.003	0.133 ± 0.003	$+0.008 \pm 0.003$	0.985 ± 0.001	0.982 ± 0.001
S_{NH4}	g N m^{-3}	LightGBM	0.115 ± 0.005	0.128 ± 0.005	$+0.013 \pm 0.001$	0.987 ± 0.001	0.984 ± 0.001
S_{NH4}	g N m^{-3}	CatBoost	0.106 ± 0.003	0.113 ± 0.004	$+0.007 \pm 0.002$	0.989 ± 0.001	0.987 ± 0.001
S_{NH4}	g N m^{-3}	AdaBoost	0.423 ± 0.009	0.471 ± 0.018	$+0.048 \pm 0.011$	0.821 ± 0.003	0.778 ± 0.008
S_{NH4}	g N m^{-3}	Random Forest	0.718 ± 0.023	0.522 ± 0.016	-0.196 ± 0.014	0.484 ± 0.011	0.727 ± 0.014
S_{NH4}	g N m^{-3}	Extra Trees	0.434 ± 0.019	0.308 ± 0.010	-0.127 ± 0.015	0.811 ± 0.009	0.905 ± 0.006
S_{NH4}	g N m^{-3}	SVR	0.222 ± 0.005	0.213 ± 0.005	-0.010 ± 0.005	0.950 ± 0.002	0.955 ± 0.001
S_{NH4}	g N m^{-3}	k-NN	0.566 ± 0.016	0.490 ± 0.009	-0.076 ± 0.009	0.680 ± 0.004	0.759 ± 0.007
S_{NH4}	g N m^{-3}	PLS	0.656 ± 0.014	0.517 ± 0.015	-0.140 ± 0.011	0.569 ± 0.004	0.733 ± 0.011
S_{NH4}	g N m^{-3}	Multi-task Elastic Net	0.495 ± 0.008	0.444 ± 0.015	-0.050 ± 0.009	0.755 ± 0.005	0.802 ± 0.006
S_{NH4}	g N m^{-3}	Multi-task Lasso	0.495 ± 0.008	0.444 ± 0.015	-0.050 ± 0.009	0.755 ± 0.005	0.802 ± 0.006
S_{NH4}	g N m^{-3}	MLP	0.079 ± 0.006	0.070 ± 0.003	-0.009 ± 0.004	0.994 ± 0.001	0.995 ± 0.001
S_{NH4}	g N m^{-3}	TabNet	0.180 ± 0.038	0.158 ± 0.033	-0.022 ± 0.012	0.966 ± 0.016	0.974 ± 0.013
S_{NO2}	g N m^{-3}	XGBoost	0.483 ± 0.017	0.469 ± 0.015	-0.015 ± 0.003	0.765 ± 0.026	0.779 ± 0.022
S_{NO2}	g N m^{-3}	LightGBM	0.468 ± 0.015	0.455 ± 0.016	-0.013 ± 0.003	0.780 ± 0.017	0.792 ± 0.013
S_{NO2}	g N m^{-3}	CatBoost	0.435 ± 0.006	0.421 ± 0.006	-0.014 ± 0.001	0.810 ± 0.012	0.822 ± 0.012
S_{NO2}	g N m^{-3}	AdaBoost	0.881 ± 0.025	0.867 ± 0.024	-0.014 ± 0.005	0.221 ± 0.056	0.246 ± 0.048
S_{NO2}	g N m^{-3}	Random Forest	0.927 ± 0.035	0.903 ± 0.035	-0.025 ± 0.002	0.139 ± 0.014	0.184 ± 0.015
S_{NO2}	g N m^{-3}	Extra Trees	0.595 ± 0.024	0.581 ± 0.023	-0.014 ± 0.006	0.646 ± 0.008	0.662 ± 0.007
S_{NO2}	g N m^{-3}	SVR	0.689 ± 0.024	0.655 ± 0.023	-0.034 ± 0.003	0.524 ± 0.024	0.570 ± 0.022
S_{NO2}	g N m^{-3}	k-NN	0.810 ± 0.033	0.798 ± 0.030	-0.012 ± 0.003	0.343 ± 0.008	0.362 ± 0.011
S_{NO2}	g N m^{-3}	PLS	0.900 ± 0.040	0.885 ± 0.039	-0.016 ± 0.003	0.189 ± 0.010	0.217 ± 0.008
S_{NO2}	g N m^{-3}	Multi-task Elastic Net	0.858 ± 0.032	0.848 ± 0.033	-0.010 ± 0.001	0.264 ± 0.018	0.281 ± 0.016
S_{NO2}	g N m^{-3}	Multi-task Lasso	0.858 ± 0.032	0.848 ± 0.033	-0.010 ± 0.001	0.264 ± 0.018	0.281 ± 0.016
S_{NO2}	g N m^{-3}	MLP	0.153 ± 0.007	0.146 ± 0.008	-0.007 ± 0.002	0.976 ± 0.003	0.979 ± 0.003

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Table S2 continued

Component	Unit	Surrogate	Raw nRMSE$_j$	Projected nRMSE$_j$	ΔnRMSE$_j$	Raw R_j^2	Projected R_j^2
S_{NO2}	g N m^{-3}	TabNet	0.317 ± 0.051	0.304 ± 0.055	-0.013 ± 0.008	0.898 ± 0.028	0.906 ± 0.029
S_{NO3}	g N m^{-3}	XGBoost	0.192 ± 0.008	0.188 ± 0.006	-0.005 ± 0.006	0.963 ± 0.004	0.965 ± 0.002
S_{NO3}	g N m^{-3}	LightGBM	0.189 ± 0.013	0.184 ± 0.009	-0.005 ± 0.004	0.964 ± 0.006	0.966 ± 0.004
S_{NO3}	g N m^{-3}	CatBoost	0.174 ± 0.005	0.168 ± 0.005	-0.006 ± 0.003	0.970 ± 0.002	0.972 ± 0.002
S_{NO3}	g N m^{-3}	AdaBoost	0.483 ± 0.004	0.499 ± 0.012	$+0.016 \pm 0.010$	0.767 ± 0.008	0.751 ± 0.008
S_{NO3}	g N m^{-3}	Random Forest	0.691 ± 0.013	0.653 ± 0.008	-0.038 ± 0.015	0.522 ± 0.008	0.573 ± 0.020
S_{NO3}	g N m^{-3}	Extra Trees	0.379 ± 0.005	0.368 ± 0.014	-0.010 ± 0.012	0.856 ± 0.004	0.864 ± 0.013
S_{NO3}	g N m^{-3}	SVR	0.303 ± 0.006	0.278 ± 0.008	-0.025 ± 0.004	0.908 ± 0.005	0.922 ± 0.004
S_{NO3}	g N m^{-3}	k-NN	0.581 ± 0.014	0.544 ± 0.011	-0.036 ± 0.005	0.663 ± 0.005	0.704 ± 0.006
S_{NO3}	g N m^{-3}	PLS	0.600 ± 0.015	0.459 ± 0.005	-0.141 ± 0.014	0.640 ± 0.008	0.789 ± 0.009
S_{NO3}	g N m^{-3}	Multi-task Elastic Net	0.544 ± 0.013	0.478 ± 0.011	-0.066 ± 0.004	0.704 ± 0.006	0.771 ± 0.002
S_{NO3}	g N m^{-3}	Multi-task Lasso	0.544 ± 0.013	0.478 ± 0.011	-0.066 ± 0.004	0.704 ± 0.006	0.771 ± 0.002
S_{NO3}	g N m^{-3}	MLP	0.081 ± 0.006	0.074 ± 0.004	-0.007 ± 0.003	0.993 ± 0.001	0.994 ± 0.001
S_{NO3}	g N m^{-3}	TabNet	0.215 ± 0.027	0.173 ± 0.020	-0.043 ± 0.015	0.953 ± 0.013	0.970 ± 0.008
S_{N2}	g N m^{-3}	XGBoost	0.387 ± 0.015	0.371 ± 0.014	-0.016 ± 0.005	0.850 ± 0.009	0.862 ± 0.007
S_{N2}	g N m^{-3}	LightGBM	0.395 ± 0.014	0.372 ± 0.016	-0.023 ± 0.006	0.844 ± 0.009	0.862 ± 0.009
S_{N2}	g N m^{-3}	CatBoost	0.351 ± 0.017	0.339 ± 0.016	-0.012 ± 0.003	0.877 ± 0.005	0.885 ± 0.005
S_{N2}	g N m^{-3}	AdaBoost	0.702 ± 0.017	0.680 ± 0.022	-0.022 ± 0.010	0.507 ± 0.014	0.538 ± 0.006
S_{N2}	g N m^{-3}	Random Forest	0.848 ± 0.022	0.797 ± 0.021	-0.050 ± 0.004	0.281 ± 0.013	0.364 ± 0.011
S_{N2}	g N m^{-3}	Extra Trees	0.559 ± 0.014	0.542 ± 0.013	-0.017 ± 0.005	0.687 ± 0.009	0.706 ± 0.010
S_{N2}	g N m^{-3}	SVR	0.554 ± 0.011	0.519 ± 0.013	-0.035 ± 0.004	0.692 ± 0.015	0.730 ± 0.010
S_{N2}	g N m^{-3}	k-NN	0.703 ± 0.023	0.683 ± 0.022	-0.020 ± 0.003	0.505 ± 0.014	0.533 ± 0.015
S_{N2}	g N m^{-3}	PLS	0.785 ± 0.025	0.764 ± 0.025	-0.022 ± 0.006	0.382 ± 0.025	0.417 ± 0.016
S_{N2}	g N m^{-3}	Multi-task Elastic Net	0.719 ± 0.024	0.694 ± 0.025	-0.024 ± 0.003	0.483 ± 0.020	0.517 ± 0.018
S_{N2}	g N m^{-3}	Multi-task Lasso	0.719 ± 0.024	0.694 ± 0.025	-0.024 ± 0.003	0.483 ± 0.020	0.517 ± 0.018
S_{N2}	g N m^{-3}	MLP	0.130 ± 0.011	0.129 ± 0.011	-0.001 ± 0.001	0.983 ± 0.003	0.983 ± 0.003
S_{N2}	g N m^{-3}	TabNet	0.301 ± 0.026	0.289 ± 0.018	-0.011 ± 0.019	0.909 ± 0.016	0.916 ± 0.011
S_{PO4}	g P m^{-3}	XGBoost	0.287 ± 0.005	0.248 ± 0.009	-0.039 ± 0.005	0.917 ± 0.003	0.938 ± 0.004
S_{PO4}	g P m^{-3}	LightGBM	0.275 ± 0.006	0.241 ± 0.007	-0.034 ± 0.003	0.925 ± 0.003	0.942 ± 0.003
S_{PO4}	g P m^{-3}	CatBoost	0.246 ± 0.004	0.224 ± 0.004	-0.022 ± 0.002	0.939 ± 0.003	0.950 ± 0.003
S_{PO4}	g P m^{-3}	AdaBoost	0.604 ± 0.007	0.524 ± 0.007	-0.080 ± 0.004	0.636 ± 0.007	0.726 ± 0.007
S_{PO4}	g P m^{-3}	Random Forest	0.814 ± 0.010	0.527 ± 0.012	-0.288 ± 0.011	0.337 ± 0.009	0.723 ± 0.012
S_{PO4}	g P m^{-3}	Extra Trees	0.663 ± 0.011	0.418 ± 0.006	-0.245 ± 0.011	0.560 ± 0.012	0.825 ± 0.005
S_{PO4}	g P m^{-3}	SVR	0.324 ± 0.009	0.304 ± 0.007	-0.020 ± 0.003	0.895 ± 0.006	0.907 ± 0.005
S_{PO4}	g P m^{-3}	k-NN	0.587 ± 0.012	0.495 ± 0.007	-0.092 ± 0.006	0.656 ± 0.009	0.755 ± 0.004
S_{PO4}	g P m^{-3}	PLS	0.637 ± 0.014	0.501 ± 0.005	-0.136 ± 0.014	0.593 ± 0.016	0.748 ± 0.007
S_{PO4}	g P m^{-3}	Multi-task Elastic Net	0.503 ± 0.008	0.482 ± 0.008	-0.021 ± 0.003	0.747 ± 0.009	0.767 ± 0.009
S_{PO4}	g P m^{-3}	Multi-task Lasso	0.503 ± 0.008	0.482 ± 0.008	-0.021 ± 0.003	0.747 ± 0.009	0.767 ± 0.009
S_{PO4}	g P m^{-3}	MLP	0.130 ± 0.011	0.118 ± 0.012	-0.013 ± 0.002	0.983 ± 0.003	0.986 ± 0.003
S_{PO4}	g P m^{-3}	TabNet	0.272 ± 0.021	0.229 ± 0.031	-0.043 ± 0.013	0.926 ± 0.012	0.947 ± 0.014
S_I	g COD m^{-3}	XGBoost	0.014 ± 0.004	0.000 ± 0.000	-0.014 ± 0.004	1.000 ± 0.000	1.000 ± 0.000
S_I	g COD m^{-3}	LightGBM	0.004 ± 0.000	0.000 ± 0.000	-0.004 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
S_I	g COD m^{-3}	CatBoost	0.007 ± 0.001	0.000 ± 0.000	-0.007 ± 0.001	1.000 ± 0.000	1.000 ± 0.000
S_I	g COD m^{-3}	AdaBoost	0.082 ± 0.010	0.000 ± 0.000	-0.082 ± 0.010	0.993 ± 0.002	1.000 ± 0.000
S_I	g COD m^{-3}	Random Forest	0.451 ± 0.017	0.000 ± 0.000	-0.451 ± 0.017	0.796 ± 0.008	1.000 ± 0.000
S_I	g COD m^{-3}	Extra Trees	0.485 ± 0.018	0.000 ± 0.000	-0.485 ± 0.018	0.765 ± 0.009	1.000 ± 0.000
S_I	g COD m^{-3}	SVR	0.073 ± 0.002	0.000 ± 0.000	-0.073 ± 0.002	0.995 ± 0.000	1.000 ± 0.000
S_I	g COD m^{-3}	k-NN	0.471 ± 0.009	0.000 ± 0.000	-0.471 ± 0.009	0.778 ± 0.003	1.000 ± 0.000
S_I	g COD m^{-3}	PLS	0.260 ± 0.033	0.000 ± 0.000	-0.260 ± 0.033	0.931 ± 0.020	1.000 ± 0.000
S_I	g COD m^{-3}	Multi-task Elastic Net	0.003 ± 0.001	0.000 ± 0.000	-0.003 ± 0.001	1.000 ± 0.000	1.000 ± 0.000
S_I	g COD m^{-3}	Multi-task Lasso	0.003 ± 0.001	0.000 ± 0.000	-0.003 ± 0.001	1.000 ± 0.000	1.000 ± 0.000
S_I	g COD m^{-3}	MLP	0.059 ± 0.010	0.000 ± 0.000	-0.059 ± 0.010	0.996 ± 0.001	1.000 ± 0.000
S_I	g COD m^{-3}	TabNet	0.196 ± 0.056	0.000 ± 0.000	-0.196 ± 0.056	0.959 ± 0.026	1.000 ± 0.000
S_{ALK}	mol m^{-3}	XGBoost	0.013 ± 0.002	0.004 ± 0.000	-0.009 ± 0.002	1.000 ± 0.000	1.000 ± 0.000
S_{ALK}	mol m^{-3}	LightGBM	0.010 ± 0.000	0.004 ± 0.000	-0.005 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
S_{ALK}	mol m^{-3}	CatBoost	0.010 ± 0.001	0.004 ± 0.000	-0.007 ± 0.001	1.000 ± 0.000	1.000 ± 0.000
S_{ALK}	mol m^{-3}	AdaBoost	0.076 ± 0.010	0.016 ± 0.001	-0.059 ± 0.010	0.994 ± 0.001	1.000 ± 0.000
S_{ALK}	mol m^{-3}	Random Forest	0.196 ± 0.005	0.017 ± 0.000	-0.178 ± 0.004	0.962 ± 0.002	1.000 ± 0.000
S_{ALK}	mol m^{-3}	Extra Trees	0.498 ± 0.018	0.010 ± 0.000	-0.488 ± 0.018	0.752 ± 0.009	1.000 ± 0.000
S_{ALK}	mol m^{-3}	SVR	0.073 ± 0.003	0.007 ± 0.000	-0.066 ± 0.003	0.995 ± 0.000	1.000 ± 0.000
S_{ALK}	mol m^{-3}	k-NN	0.470 ± 0.007	0.017 ± 0.000	-0.453 ± 0.007	0.779 ± 0.003	1.000 ± 0.000
S_{ALK}	mol m^{-3}	PLS	0.449 ± 0.073	0.014 ± 0.000	-0.435 ± 0.073	0.792 ± 0.063	1.000 ± 0.000
S_{ALK}	mol m^{-3}	Multi-task Elastic Net	0.017 ± 0.000	0.014 ± 0.000	-0.002 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
S_{ALK}	mol m^{-3}	Multi-task Lasso	0.017 ± 0.000	0.014 ± 0.000	-0.002 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
S_{ALK}	mol m^{-3}	MLP	0.063 ± 0.014	0.002 ± 0.000	-0.061 ± 0.014	0.996 ± 0.002	1.000 ± 0.000
S_{ALK}	mol m^{-3}	TabNet	0.156 ± 0.026	0.005 ± 0.001	-0.151 ± 0.026	0.975 ± 0.008	1.000 ± 0.000
X_I	g COD m^{-3}	XGBoost	0.027 ± 0.004	0.043 ± 0.004	$+0.016 \pm 0.002$	0.999 ± 0.000	0.998 ± 0.000
X_I	g COD m^{-3}	LightGBM	0.018 ± 0.001	0.037 ± 0.002	$+0.019 \pm 0.001$	1.000 ± 0.000	0.999 ± 0.000
X_I	g COD m^{-3}	CatBoost	0.014 ± 0.001	0.030 ± 0.003	$+0.015 \pm 0.002$	1.000 ± 0.000	0.999 ± 0.000
X_I	g COD m^{-3}	AdaBoost	0.131 ± 0.002	0.160 ± 0.004	$+0.029 \pm 0.002$	0.983 ± 0.001	0.974 ± 0.001
X_I	g COD m^{-3}	Random Forest	0.357 ± 0.004	0.376 ± 0.004	$+0.019 \pm 0.002$	0.872 ± 0.003	0.858 ± 0.002
X_I	g COD m^{-3}	Extra Trees	0.502 ± 0.009	0.508 ± 0.011	$+0.006 \pm 0.002$	0.748 ± 0.005	0.742 ± 0.006
X_I	g COD m^{-3}	SVR	0.074 ± 0.002	0.083 ± 0.003	$+0.010 \pm 0.001$	0.995 ± 0.000	0.993 ± 0.000
X_I	g COD m^{-3}	k-NN	0.468 ± 0.008	0.476 ± 0.009	$+0.008 \pm 0.003$	0.781 ± 0.004	0.774 ± 0.006
X_I	g COD m^{-3}	PLS	0.516 ± 0.045	0.511 ± 0.039	-0.005 ± 0.008	0.731 ± 0.049	0.737 ± 0.042
X_I	g COD m^{-3}	Multi-task Elastic Net	0.029 ± 0.001	0.044 ± 0.002	$+0.015 \pm 0.002$	0.999 ± 0.000	0.998 ± 0.000
X_I	g COD m^{-3}	Multi-task Lasso	0.029 ± 0.001	0.044 ± 0.002	$+0.016 \pm 0.002$	0.999 ± 0.000	0.998 ± 0.000
X_I	g COD m^{-3}	MLP	0.061 ± 0.009	0.065 ± 0.009	$+0.003 \pm 0.001$	0.996 ± 0.001	0.996 ± 0.001
X_I	g COD m^{-3}	TabNet	0.150 ± 0.024	0.156 ± 0.025	$+0.007 \pm 0.007$	0.977 ± 0.007	0.975 ± 0.008
X_S	g COD m^{-3}	XGBoost	0.113 ± 0.007	0.113 ± 0.007	$+0.000 \pm 0.000$	0.987 ± 0.001	0.987 ± 0.001

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Table S2 continued

Component	Unit	Surrogate	Raw nRMSE$_j$	Projected nRMSE$_j$	ΔnRMSE$_j$	Raw R_j^2	Projected R_j^2
X_S	g COD m^{-3}	LightGBM	0.101 ± 0.004	0.101 ± 0.004	$+0.000 \pm 0.000$	0.990 ± 0.001	0.990 ± 0.001
X_S	g COD m^{-3}	CatBoost	0.086 ± 0.003	0.086 ± 0.003	-0.000 ± 0.000	0.993 ± 0.001	0.993 ± 0.001
X_S	g COD m^{-3}	AdaBoost	0.375 ± 0.018	0.388 ± 0.012	$+0.014 \pm 0.007$	0.860 ± 0.004	0.849 ± 0.002
X_S	g COD m^{-3}	Random Forest	0.304 ± 0.012	0.371 ± 0.008	$+0.067 \pm 0.014$	0.907 ± 0.004	0.862 ± 0.012
X_S	g COD m^{-3}	Extra Trees	0.326 ± 0.018	0.373 ± 0.005	$+0.047 \pm 0.013$	0.894 ± 0.004	0.860 ± 0.006
X_S	g COD m^{-3}	SVR	0.225 ± 0.009	0.225 ± 0.009	-0.000 ± 0.000	0.949 ± 0.002	0.949 ± 0.002
X_S	g COD m^{-3}	k-NN	0.607 ± 0.032	0.616 ± 0.028	$+0.008 \pm 0.004$	0.631 ± 0.013	0.621 ± 0.009
X_S	g COD m^{-3}	PLS	0.708 ± 0.024	0.713 ± 0.019	$+0.005 \pm 0.008$	0.498 ± 0.004	0.490 ± 0.015
X_S	g COD m^{-3}	Multi-task Elastic Net	0.572 ± 0.019	0.525 ± 0.020	-0.047 ± 0.003	0.673 ± 0.004	0.724 ± 0.002
X_S	g COD m^{-3}	Multi-task Lasso	0.572 ± 0.019	0.525 ± 0.020	-0.047 ± 0.003	0.673 ± 0.004	0.724 ± 0.002
X_S	g COD m^{-3}	MLP	0.088 ± 0.007	0.088 ± 0.007	-0.000 ± 0.000	0.992 ± 0.002	0.992 ± 0.002
X_S	g COD m^{-3}	TabNet	0.195 ± 0.035	0.199 ± 0.038	$+0.004 \pm 0.011$	0.960 ± 0.017	0.958 ± 0.018
X_H	g COD m^{-3}	XGBoost	0.129 ± 0.006	0.272 ± 0.012	$+0.142 \pm 0.008$	0.983 ± 0.001	0.926 ± 0.005
X_H	g COD m^{-3}	LightGBM	0.123 ± 0.004	0.269 ± 0.008	$+0.146 \pm 0.007$	0.985 ± 0.001	0.928 ± 0.002
X_H	g COD m^{-3}	CatBoost	0.093 ± 0.002	0.201 ± 0.008	$+0.107 \pm 0.007$	0.991 ± 0.001	0.960 ± 0.002
X_H	g COD m^{-3}	AdaBoost	0.509 ± 0.009	0.955 ± 0.028	$+0.445 \pm 0.030$	0.740 ± 0.013	0.089 ± 0.025
X_H	g COD m^{-3}	Random Forest	0.410 ± 0.012	1.319 ± 0.017	$+0.909 \pm 0.008$	0.832 ± 0.002	-0.742 ± 0.059
X_H	g COD m^{-3}	Extra Trees	0.693 ± 0.018	0.914 ± 0.017	$+0.220 \pm 0.009$	0.519 ± 0.006	0.165 ± 0.023
X_H	g COD m^{-3}	SVR	0.174 ± 0.005	0.285 ± 0.013	$+0.111 \pm 0.014$	0.970 ± 0.001	0.918 ± 0.007
X_H	g COD m^{-3}	k-NN	0.542 ± 0.018	0.889 ± 0.014	$+0.346 \pm 0.008$	0.706 ± 0.008	0.210 ± 0.018
X_H	g COD m^{-3}	PLS	0.859 ± 0.019	0.987 ± 0.024	$+0.128 \pm 0.019$	0.262 ± 0.011	0.025 ± 0.051
X_H	g COD m^{-3}	Multi-task Elastic Net	0.420 ± 0.011	0.471 ± 0.012	$+0.051 \pm 0.008$	0.823 ± 0.005	0.778 ± 0.005
X_H	g COD m^{-3}	Multi-task Lasso	0.420 ± 0.011	0.471 ± 0.012	$+0.051 \pm 0.008$	0.823 ± 0.005	0.778 ± 0.005
X_H	g COD m^{-3}	MLP	0.092 ± 0.011	0.157 ± 0.024	$+0.065 \pm 0.019$	0.991 ± 0.002	0.975 ± 0.008
X_H	g COD m^{-3}	TabNet	0.208 ± 0.051	0.398 ± 0.035	$+0.190 \pm 0.036$	0.954 ± 0.025	0.840 ± 0.033
X_{PAO}	g COD m^{-3}	XGBoost	0.109 ± 0.005	0.113 ± 0.004	$+0.004 \pm 0.002$	0.988 ± 0.001	0.987 ± 0.001
X_{PAO}	g COD m^{-3}	LightGBM	0.104 ± 0.004	0.107 ± 0.004	$+0.003 \pm 0.002$	0.989 ± 0.001	0.989 ± 0.001
X_{PAO}	g COD m^{-3}	CatBoost	0.093 ± 0.006	0.096 ± 0.005	$+0.003 \pm 0.001$	0.991 ± 0.001	0.991 ± 0.001
X_{PAO}	g COD m^{-3}	AdaBoost	0.220 ± 0.002	0.239 ± 0.003	$+0.018 \pm 0.001$	0.951 ± 0.001	0.943 ± 0.001
X_{PAO}	g COD m^{-3}	Random Forest	0.609 ± 0.011	0.611 ± 0.008	$+0.002 \pm 0.003$	0.629 ± 0.003	0.626 ± 0.001
X_{PAO}	g COD m^{-3}	Extra Trees	0.402 ± 0.010	0.413 ± 0.010	$+0.010 \pm 0.003$	0.838 ± 0.004	0.830 ± 0.004
X_{PAO}	g COD m^{-3}	SVR	0.140 ± 0.004	0.144 ± 0.005	$+0.004 \pm 0.002$	0.980 ± 0.001	0.979 ± 0.001
X_{PAO}	g COD m^{-3}	k-NN	0.478 ± 0.009	0.483 ± 0.008	$+0.005 \pm 0.002$	0.772 ± 0.003	0.767 ± 0.003
X_{PAO}	g COD m^{-3}	PLS	0.611 ± 0.023	0.605 ± 0.022	-0.006 ± 0.005	0.627 ± 0.025	0.634 ± 0.023
X_{PAO}	g COD m^{-3}	Multi-task Elastic Net	0.178 ± 0.004	0.175 ± 0.004	-0.002 ± 0.001	0.968 ± 0.001	0.969 ± 0.001
X_{PAO}	g COD m^{-3}	Multi-task Lasso	0.178 ± 0.004	0.175 ± 0.004	-0.002 ± 0.001	0.968 ± 0.001	0.969 ± 0.001
X_{PAO}	g COD m^{-3}	MLP	0.091 ± 0.005	0.093 ± 0.006	$+0.002 \pm 0.001$	0.992 ± 0.001	0.991 ± 0.001
X_{PAO}	g COD m^{-3}	TabNet	0.203 ± 0.037	0.202 ± 0.031	-0.001 ± 0.008	0.957 ± 0.018	0.958 ± 0.015
X_{PP}	g P m^{-3}	XGBoost	0.300 ± 0.012	0.312 ± 0.011	$+0.012 \pm 0.004$	0.910 ± 0.009	0.903 ± 0.008
X_{PP}	g P m^{-3}	LightGBM	0.297 ± 0.009	0.306 ± 0.009	$+0.009 \pm 0.005$	0.912 ± 0.005	0.906 ± 0.006
X_{PP}	g P m^{-3}	CatBoost	0.286 ± 0.007	0.282 ± 0.006	-0.004 ± 0.004	0.918 ± 0.006	0.920 ± 0.005
X_{PP}	g P m^{-3}	AdaBoost	0.628 ± 0.011	0.643 ± 0.008	$+0.015 \pm 0.003$	0.606 ± 0.006	0.587 ± 0.007
X_{PP}	g P m^{-3}	Random Forest	0.729 ± 0.009	0.673 ± 0.009	-0.056 ± 0.010	0.467 ± 0.013	0.546 ± 0.021
X_{PP}	g P m^{-3}	Extra Trees	0.614 ± 0.009	0.533 ± 0.006	-0.081 ± 0.009	0.623 ± 0.013	0.716 ± 0.013
X_{PP}	g P m^{-3}	SVR	0.392 ± 0.011	0.378 ± 0.009	-0.014 ± 0.003	0.846 ± 0.008	0.857 ± 0.007
X_{PP}	g P m^{-3}	k-NN	0.624 ± 0.009	0.602 ± 0.008	-0.021 ± 0.009	0.611 ± 0.008	0.637 ± 0.012
X_{PP}	g P m^{-3}	PLS	0.770 ± 0.019	0.620 ± 0.006	-0.150 ± 0.015	0.407 ± 0.014	0.615 ± 0.010
X_{PP}	g P m^{-3}	Multi-task Elastic Net	0.588 ± 0.011	0.582 ± 0.010	-0.006 ± 0.003	0.653 ± 0.011	0.661 ± 0.010
X_{PP}	g P m^{-3}	Multi-task Lasso	0.588 ± 0.011	0.582 ± 0.010	-0.006 ± 0.003	0.653 ± 0.011	0.661 ± 0.010
X_{PP}	g P m^{-3}	MLP	0.152 ± 0.014	0.143 ± 0.013	-0.009 ± 0.003	0.977 ± 0.005	0.979 ± 0.004
X_{PP}	g P m^{-3}	TabNet	0.302 ± 0.044	0.288 ± 0.035	-0.014 ± 0.010	0.907 ± 0.028	0.916 ± 0.022
X_{PHA}	g COD m^{-3}	XGBoost	0.311 ± 0.015	0.311 ± 0.015	-0.001 ± 0.000	0.903 ± 0.006	0.903 ± 0.006
X_{PHA}	g COD m^{-3}	LightGBM	0.303 ± 0.011	0.302 ± 0.011	-0.001 ± 0.000	0.908 ± 0.006	0.909 ± 0.006
X_{PHA}	g COD m^{-3}	CatBoost	0.295 ± 0.009	0.293 ± 0.009	-0.002 ± 0.000	0.913 ± 0.004	0.914 ± 0.004
X_{PHA}	g COD m^{-3}	AdaBoost	0.659 ± 0.023	0.660 ± 0.023	$+0.001 \pm 0.000$	0.565 ± 0.025	0.564 ± 0.026
X_{PHA}	g COD m^{-3}	Random Forest	0.714 ± 0.015	0.715 ± 0.015	$+0.001 \pm 0.001$	0.490 ± 0.009	0.489 ± 0.009
X_{PHA}	g COD m^{-3}	Extra Trees	0.619 ± 0.016	0.620 ± 0.016	$+0.001 \pm 0.001$	0.617 ± 0.010	0.615 ± 0.011
X_{PHA}	g COD m^{-3}	SVR	0.415 ± 0.011	0.406 ± 0.011	-0.010 ± 0.001	0.827 ± 0.005	0.835 ± 0.005
X_{PHA}	g COD m^{-3}	k-NN	0.636 ± 0.014	0.636 ± 0.013	$+0.000 \pm 0.000$	0.596 ± 0.006	0.596 ± 0.006
X_{PHA}	g COD m^{-3}	PLS	0.741 ± 0.014	0.733 ± 0.015	-0.008 ± 0.001	0.450 ± 0.008	0.462 ± 0.009
X_{PHA}	g COD m^{-3}	Multi-task Elastic Net	0.600 ± 0.016	0.579 ± 0.017	-0.021 ± 0.005	0.640 ± 0.011	0.665 ± 0.014
X_{PHA}	g COD m^{-3}	Multi-task Lasso	0.600 ± 0.016	0.579 ± 0.017	-0.021 ± 0.004	0.640 ± 0.011	0.665 ± 0.014
X_{PHA}	g COD m^{-3}	MLP	0.166 ± 0.011	0.164 ± 0.011	-0.003 ± 0.001	0.972 ± 0.004	0.973 ± 0.004
X_{PHA}	g COD m^{-3}	TabNet	0.350 ± 0.092	0.344 ± 0.095	-0.006 ± 0.004	0.869 ± 0.078	0.872 ± 0.079
X_{AOB}	g COD m^{-3}	XGBoost	0.110 ± 0.004	0.110 ± 0.004	-0.001 ± 0.000	0.988 ± 0.001	0.988 ± 0.001
X_{AOB}	g COD m^{-3}	LightGBM	0.105 ± 0.004	0.104 ± 0.003	-0.001 ± 0.000	0.989 ± 0.001	0.989 ± 0.001
X_{AOB}	g COD m^{-3}	CatBoost	0.084 ± 0.002	0.084 ± 0.002	-0.000 ± 0.000	0.993 ± 0.000	0.993 ± 0.000
X_{AOB}	g COD m^{-3}	AdaBoost	0.448 ± 0.004	0.446 ± 0.004	-0.002 ± 0.000	0.799 ± 0.004	0.801 ± 0.004
X_{AOB}	g COD m^{-3}	Random Forest	0.892 ± 0.009	0.884 ± 0.008	-0.009 ± 0.001	0.204 ± 0.010	0.219 ± 0.009
X_{AOB}	g COD m^{-3}	Extra Trees	0.557 ± 0.005	0.556 ± 0.005	-0.001 ± 0.001	0.690 ± 0.005	0.691 ± 0.005
X_{AOB}	g COD m^{-3}	SVR	0.165 ± 0.002	0.164 ± 0.002	-0.001 ± 0.000	0.973 ± 0.001	0.973 ± 0.001
X_{AOB}	g COD m^{-3}	k-NN	0.508 ± 0.005	0.505 ± 0.004	-0.002 ± 0.000	0.742 ± 0.004	0.745 ± 0.004
X_{AOB}	g COD m^{-3}	PLS	0.644 ± 0.018	0.647 ± 0.018	$+0.003 \pm 0.001$	0.585 ± 0.021	0.582 ± 0.022
X_{AOB}	g COD m^{-3}	Multi-task Elastic Net	0.335 ± 0.004	0.332 ± 0.004	-0.003 ± 0.000	0.888 ± 0.003	0.890 ± 0.003
X_{AOB}	g COD m^{-3}	Multi-task Lasso	0.335 ± 0.004	0.332 ± 0.004	-0.003 ± 0.000	0.888 ± 0.003	0.890 ± 0.003
X_{AOB}	g COD m^{-3}	MLP	0.080 ± 0.007	0.080 ± 0.007	-0.000 ± 0.000	0.993 ± 0.001	0.993 ± 0.001
X_{AOB}	g COD m^{-3}	TabNet	0.164 ± 0.035	0.163 ± 0.035	-0.001 ± 0.001	0.972 ± 0.013	0.972 ± 0.013
X_{NOB}	g COD m^{-3}	XGBoost	0.120 ± 0.003	0.120 ± 0.003	-0.000 ± 0.000	0.986 ± 0.001	0.986 ± 0.001
X_{NOB}	g COD m^{-3}	LightGBM	0.116 ± 0.003	0.116 ± 0.003	-0.000 ± 0.000	0.987 ± 0.001	0.987 ± 0.001
X_{NOB}	g COD m^{-3}	CatBoost	0.100 ± 0.004	0.100 ± 0.004	-0.000 ± 0.000	0.990 ± 0.001	0.990 ± 0.001

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Table S2 continued

Component	Unit	Surrogate	Raw $nRMSE_j$	Projected $nRMSE_j$	$\Delta nRMSE_j$	Raw R_j^2	Projected R_j^2
X_{NOB}	g COD m^{-3}	AdaBoost	0.314 ± 0.003	0.311 ± 0.003	-0.003 ± 0.000	0.901 ± 0.003	0.903 ± 0.003
X_{NOB}	g COD m^{-3}	Random Forest	0.925 ± 0.010	0.920 ± 0.011	-0.005 ± 0.001	0.144 ± 0.019	0.154 ± 0.018
X_{NOB}	g COD m^{-3}	Extra Trees	0.511 ± 0.009	0.511 ± 0.010	-0.000 ± 0.001	0.738 ± 0.009	0.739 ± 0.009
X_{NOB}	g COD m^{-3}	SVR	0.170 ± 0.003	0.170 ± 0.003	-0.000 ± 0.000	0.971 ± 0.001	0.971 ± 0.001
X_{NOB}	g COD m^{-3}	k-NN	0.501 ± 0.004	0.500 ± 0.004	-0.001 ± 0.001	0.749 ± 0.003	0.750 ± 0.002
X_{NOB}	g COD m^{-3}	PLS	0.846 ± 0.017	0.839 ± 0.018	-0.007 ± 0.001	0.284 ± 0.020	0.295 ± 0.021
X_{NOB}	g COD m^{-3}	Multi-task Elastic Net	0.246 ± 0.002	0.244 ± 0.002	-0.002 ± 0.000	0.940 ± 0.002	0.941 ± 0.002
X_{NOB}	g COD m^{-3}	Multi-task Lasso	0.246 ± 0.002	0.244 ± 0.002	-0.002 ± 0.000	0.940 ± 0.002	0.941 ± 0.002
X_{NOB}	g COD m^{-3}	MLP	0.089 ± 0.012	0.089 ± 0.012	-0.000 ± 0.000	0.992 ± 0.002	0.992 ± 0.002
X_{NOB}	g COD m^{-3}	TabNet	0.221 ± 0.131	0.220 ± 0.132	-0.001 ± 0.001	0.938 ± 0.078	0.938 ± 0.078
X_{MeP}	g TSS m^{-3}	XGBoost	0.220 ± 0.008	0.198 ± 0.004	-0.022 ± 0.005	0.951 ± 0.005	0.961 ± 0.002
X_{MeP}	g TSS m^{-3}	LightGBM	0.205 ± 0.009	0.187 ± 0.005	-0.019 ± 0.005	0.958 ± 0.005	0.965 ± 0.003
X_{MeP}	g TSS m^{-3}	CatBoost	0.195 ± 0.003	0.185 ± 0.004	-0.010 ± 0.003	0.962 ± 0.002	0.966 ± 0.002
X_{MeP}	g TSS m^{-3}	AdaBoost	0.529 ± 0.010	0.395 ± 0.007	-0.134 ± 0.009	0.720 ± 0.016	0.844 ± 0.008
X_{MeP}	g TSS m^{-3}	Random Forest	0.937 ± 0.015	0.489 ± 0.009	-0.449 ± 0.019	0.121 ± 0.005	0.761 ± 0.013
X_{MeP}	g TSS m^{-3}	Extra Trees	0.803 ± 0.014	0.357 ± 0.011	-0.446 ± 0.017	0.355 ± 0.009	0.872 ± 0.009
X_{MeP}	g TSS m^{-3}	SVR	0.220 ± 0.003	0.203 ± 0.003	-0.017 ± 0.003	0.951 ± 0.002	0.959 ± 0.002
X_{MeP}	g TSS m^{-3}	k-NN	0.522 ± 0.008	0.334 ± 0.004	-0.188 ± 0.007	0.727 ± 0.006	0.889 ± 0.003
X_{MeP}	g TSS m^{-3}	PLS	0.868 ± 0.023	0.383 ± 0.010	-0.485 ± 0.028	0.245 ± 0.030	0.853 ± 0.010
X_{MeP}	g TSS m^{-3}	Multi-task Elastic Net	0.379 ± 0.009	0.353 ± 0.010	-0.026 ± 0.001	0.856 ± 0.010	0.875 ± 0.009
X_{MeP}	g TSS m^{-3}	Multi-task Lasso	0.379 ± 0.009	0.353 ± 0.010	-0.026 ± 0.001	0.856 ± 0.010	0.875 ± 0.009
X_{MeP}	g TSS m^{-3}	MLP	0.125 ± 0.008	0.109 ± 0.005	-0.016 ± 0.004	0.984 ± 0.002	0.988 ± 0.001
X_{MeP}	g TSS m^{-3}	TabNet	0.255 ± 0.066	0.178 ± 0.020	-0.077 ± 0.058	0.931 ± 0.037	0.968 ± 0.007
X_{MeOH}	g TSS m^{-3}	XGBoost	0.337 ± 0.008	0.332 ± 0.008	-0.005 ± 0.005	0.886 ± 0.008	0.890 ± 0.007
X_{MeOH}	g TSS m^{-3}	LightGBM	0.317 ± 0.010	0.312 ± 0.009	-0.005 ± 0.005	0.899 ± 0.009	0.902 ± 0.008
X_{MeOH}	g TSS m^{-3}	CatBoost	0.316 ± 0.009	0.309 ± 0.008	-0.007 ± 0.003	0.900 ± 0.005	0.904 ± 0.004
X_{MeOH}	g TSS m^{-3}	AdaBoost	0.675 ± 0.016	0.661 ± 0.014	-0.014 ± 0.003	0.544 ± 0.020	0.562 ± 0.021
X_{MeOH}	g TSS m^{-3}	Random Forest	0.906 ± 0.028	0.818 ± 0.022	-0.088 ± 0.006	0.179 ± 0.006	0.330 ± 0.010
X_{MeOH}	g TSS m^{-3}	Extra Trees	0.700 ± 0.026	0.598 ± 0.022	-0.102 ± 0.010	0.510 ± 0.006	0.642 ± 0.012
X_{MeOH}	g TSS m^{-3}	SVR	0.351 ± 0.006	0.340 ± 0.005	-0.012 ± 0.002	0.876 ± 0.009	0.884 ± 0.008
X_{MeOH}	g TSS m^{-3}	k-NN	0.615 ± 0.012	0.558 ± 0.008	-0.056 ± 0.006	0.622 ± 0.013	0.687 ± 0.015
X_{MeOH}	g TSS m^{-3}	PLS	0.810 ± 0.030	0.641 ± 0.021	-0.168 ± 0.014	0.344 ± 0.016	0.588 ± 0.011
X_{MeOH}	g TSS m^{-3}	Multi-task Elastic Net	0.634 ± 0.021	0.590 ± 0.021	-0.044 ± 0.001	0.598 ± 0.003	0.652 ± 0.004
X_{MeOH}	g TSS m^{-3}	Multi-task Lasso	0.634 ± 0.021	0.590 ± 0.021	-0.044 ± 0.001	0.598 ± 0.003	0.652 ± 0.004
X_{MeOH}	g TSS m^{-3}	MLP	0.189 ± 0.008	0.182 ± 0.009	-0.007 ± 0.002	0.964 ± 0.003	0.967 ± 0.004
X_{MeOH}	g TSS m^{-3}	TabNet	0.314 ± 0.048	0.298 ± 0.033	-0.016 ± 0.020	0.898 ± 0.033	0.910 ± 0.022

Table S3 reports composite-normalized RMSE for COD, TN, TP, and TSS. Projection lowered TN and TP error for every surrogate but increased COD and TSS error for every surrogate.

Table S3: Full-size in-distribution accuracy by derived water-quality composite and surrogate. Composite nRMSE is physical-unit RMSE divided by the population SD of that composite in the applicable outer-training partition. Values are five-fold mean ± sample SD and Δ is projected minus raw on paired folds.

Composite	Surrogate	Raw nRMSE_q	Projected nRMSE_q	$\Delta\mathrm{nRMSE}_q$	Raw R^2	Projected R^2
COD	XGBoost	0.105 ± 0.005	0.191 ± 0.008	+0.086 ± 0.004	0.989 ± 0.001	0.963 ± 0.003
COD	LightGBM	0.094 ± 0.006	0.188 ± 0.005	+0.094 ± 0.004	0.991 ± 0.001	0.964 ± 0.002
COD	CatBoost	0.069 ± 0.003	0.141 ± 0.007	+0.071 ± 0.005	0.995 ± 0.000	0.980 ± 0.001
COD	AdaBoost	0.396 ± 0.009	0.574 ± 0.010	+0.177 ± 0.010	0.842 ± 0.009	0.670 ± 0.012
COD	Random Forest	0.416 ± 0.018	0.872 ± 0.011	+0.456 ± 0.016	0.827 ± 0.007	0.238 ± 0.039
COD	Extra Trees	0.555 ± 0.023	0.650 ± 0.019	+0.095 ± 0.008	0.692 ± 0.010	0.577 ± 0.008
COD	SVR	0.115 ± 0.005	0.214 ± 0.007	+0.099 ± 0.003	0.987 ± 0.001	0.954 ± 0.003
COD	k-NN	0.487 ± 0.020	0.655 ± 0.020	+0.168 ± 0.004	0.762 ± 0.007	0.570 ± 0.011
COD	PLS	0.565 ± 0.035	0.732 ± 0.027	+0.167 ± 0.013	0.681 ± 0.030	0.464 ± 0.028
COD	Multi-task Elastic Net	0.202 ± 0.006	0.333 ± 0.012	+0.131 ± 0.009	0.959 ± 0.002	0.889 ± 0.004
COD	Multi-task Lasso	0.202 ± 0.006	0.333 ± 0.012	+0.131 ± 0.009	0.959 ± 0.002	0.889 ± 0.004
COD	MLP	0.067 ± 0.008	0.107 ± 0.014	+0.040 ± 0.007	0.995 ± 0.001	0.988 ± 0.003
COD	TabNet	0.188 ± 0.049	0.285 ± 0.035	+0.096 ± 0.037	0.962 ± 0.019	0.917 ± 0.025
TN	XGBoost	0.226 ± 0.009	0.180 ± 0.007	−0.045 ± 0.006	0.949 ± 0.004	0.967 ± 0.002
TN	LightGBM	0.221 ± 0.010	0.180 ± 0.007	−0.040 ± 0.005	0.951 ± 0.004	0.967 ± 0.002
TN	CatBoost	0.194 ± 0.002	0.164 ± 0.007	−0.029 ± 0.007	0.962 ± 0.001	0.973 ± 0.002
TN	AdaBoost	0.530 ± 0.007	0.330 ± 0.009	−0.200 ± 0.005	0.719 ± 0.012	0.891 ± 0.006
TN	Random Forest	0.896 ± 0.011	0.387 ± 0.008	−0.509 ± 0.012	0.197 ± 0.006	0.850 ± 0.007
TN	Extra Trees	0.510 ± 0.014	0.263 ± 0.005	−0.247 ± 0.012	0.740 ± 0.008	0.931 ± 0.002
TN	SVR	0.297 ± 0.005	0.252 ± 0.005	−0.045 ± 0.003	0.912 ± 0.004	0.936 ± 0.002
TN	k-NN	0.543 ± 0.010	0.332 ± 0.010	−0.212 ± 0.007	0.705 ± 0.004	0.890 ± 0.005
TN	PLS	0.838 ± 0.016	0.371 ± 0.011	−0.467 ± 0.009	0.298 ± 0.009	0.862 ± 0.006
TN	Multi-task Elastic Net	0.349 ± 0.011	0.337 ± 0.011	−0.012 ± 0.001	0.878 ± 0.005	0.886 ± 0.006
TN	Multi-task Lasso	0.349 ± 0.011	0.337 ± 0.011	−0.012 ± 0.001	0.878 ± 0.005	0.886 ± 0.006
TN	MLP	0.096 ± 0.011	0.063 ± 0.005	−0.033 ± 0.007	0.991 ± 0.002	0.996 ± 0.001
TN	TabNet	0.269 ± 0.047	0.141 ± 0.008	−0.128 ± 0.045	0.926 ± 0.025	0.980 ± 0.003
TP	XGBoost	0.148 ± 0.010	0.000 ± 0.000	−0.148 ± 0.010	0.978 ± 0.002	1.000 ± 0.000
TP	LightGBM	0.138 ± 0.006	0.000 ± 0.000	−0.138 ± 0.006	0.981 ± 0.001	1.000 ± 0.000
TP	CatBoost	0.108 ± 0.008	0.000 ± 0.000	−0.108 ± 0.008	0.988 ± 0.002	1.000 ± 0.000
TP	AdaBoost	0.389 ± 0.012	0.000 ± 0.000	−0.388 ± 0.012	0.849 ± 0.009	1.000 ± 0.000
TP	Random Forest	0.942 ± 0.018	0.000 ± 0.000	−0.941 ± 0.018	0.112 ± 0.005	1.000 ± 0.000
TP	Extra Trees	0.761 ± 0.020	0.000 ± 0.000	−0.761 ± 0.020	0.421 ± 0.009	1.000 ± 0.000
TP	SVR	0.075 ± 0.003	0.000 ± 0.000	−0.075 ± 0.003	0.994 ± 0.000	1.000 ± 0.000
TP	k-NN	0.464 ± 0.011	0.000 ± 0.000	−0.464 ± 0.011	0.784 ± 0.005	1.000 ± 0.000
TP	PLS	0.708 ± 0.035	0.000 ± 0.000	−0.707 ± 0.035	0.499 ± 0.033	1.000 ± 0.000
TP	Multi-task Elastic Net	0.004 ± 0.001	0.000 ± 0.000	−0.004 ± 0.001	1.000 ± 0.000	1.000 ± 0.000
TP	Multi-task Lasso	0.004 ± 0.001	0.000 ± 0.000	−0.004 ± 0.001	1.000 ± 0.000	1.000 ± 0.000
TP	MLP	0.072 ± 0.006	0.000 ± 0.000	−0.072 ± 0.006	0.995 ± 0.001	1.000 ± 0.000
TP	TabNet	0.207 ± 0.048	0.000 ± 0.000	−0.207 ± 0.048	0.955 ± 0.019	1.000 ± 0.000
TSS	XGBoost	0.127 ± 0.004	0.226 ± 0.011	+0.099 ± 0.007	0.984 ± 0.001	0.949 ± 0.004
TSS	LightGBM	0.123 ± 0.003	0.220 ± 0.008	+0.097 ± 0.006	0.985 ± 0.001	0.952 ± 0.002
TSS	CatBoost	0.097 ± 0.004	0.169 ± 0.007	+0.072 ± 0.006	0.991 ± 0.001	0.972 ± 0.000
TSS	AdaBoost	0.413 ± 0.008	0.705 ± 0.015	+0.292 ± 0.009	0.829 ± 0.007	0.502 ± 0.011
TSS	Random Forest	0.465 ± 0.015	1.044 ± 0.012	+0.579 ± 0.015	0.784 ± 0.002	−0.093 ± 0.061
TSS	Extra Trees	0.631 ± 0.024	0.761 ± 0.021	+0.129 ± 0.007	0.601 ± 0.006	0.421 ± 0.012
TSS	SVR	0.137 ± 0.004	0.227 ± 0.005	+0.090 ± 0.004	0.981 ± 0.000	0.948 ± 0.003
TSS	k-NN	0.503 ± 0.019	0.727 ± 0.019	+0.224 ± 0.005	0.747 ± 0.006	0.470 ± 0.012
TSS	PLS	0.837 ± 0.030	0.843 ± 0.030	+0.006 ± 0.021	0.299 ± 0.019	0.289 ± 0.031
TSS	Multi-task Elastic Net	0.229 ± 0.009	0.341 ± 0.010	+0.112 ± 0.012	0.947 ± 0.002	0.883 ± 0.006
TSS	Multi-task Lasso	0.229 ± 0.009	0.341 ± 0.010	+0.112 ± 0.012	0.947 ± 0.002	0.883 ± 0.006
TSS	MLP	0.087 ± 0.011	0.134 ± 0.018	+0.047 ± 0.008	0.992 ± 0.002	0.982 ± 0.004
TSS	TabNet	0.214 ± 0.045	0.337 ± 0.036	+0.123 ± 0.017	0.952 ± 0.022	0.884 ± 0.032

4 Mass-Conservation, Non-Negativity, and Positivity Backtracking

All 750,750 raw in-distribution predictions violated the declared mass-conservation tolerance. Raw non-negativity violations varied by surrogate and sample size. After projection, every in-distribution prediction satisfied mass conservation within tolerance and contained no negative concentration; the maximum projected conservation residual was 2.06×10^{-13}.

Positivity backtracking was activated for 2,489 predictions (0.332%). Random Forest, PLS, and Extra Trees accounted for 2,044 activations (82.1%) and had among the largest standardized projection displacements. Table S4 distinguishes the cumulative count from the activation rate so that the larger number of evaluated rows at high sample totals is not mistaken for a rising backtracking propensity.

Table S4: Positivity-backtracking activation by surrogate. The all-size denominator is 57,750 projections per surrogate; the final two columns describe $N = 10{,}000$.

Surrogate	All-size count	All-size rate (%)	Full-size rate (%)	Full-size mean d_c^{std}
XGBoost	4	0.007	0.000	0.300
LightGBM	3	0.005	0.000	0.301
CatBoost	12	0.021	0.000	0.242
AdaBoost	177	0.306	0.300	0.901
Random Forest	904	1.565	1.420	2.002
Extra Trees	566	0.980	0.880	1.528
SVR	17	0.029	0.010	0.363
k-NN	165	0.286	0.250	1.192
PLS	574	0.994	0.930	1.808
Multi-task Elastic Net	0	0.000	0.000	0.482
Multi-task Lasso	0	0.000	0.000	0.482
MLP	6	0.010	0.000	0.189
TabNet	61	0.106	0.040	0.545

5 Computational Performance

Fit time includes fold-local preprocessing and surrogate fitting, but excludes Optuna search, model saving, prediction, and projection. Inference timing used a deterministic 512-row batch, five warm-up calls, and 30 measured calls. Values in Table S5 are five-fold mean ± sample SD. Search duration is omitted because each fold-specific study was reused at all eleven nested sizes and therefore is not a size-specific training cost.

Training-time scaling differed much more strongly than inference latency (Figure S1). At $N = 10{,}000$, mean fold-local preprocessing and fitting time ranged from 0.024 s for Multi-task Elastic Net to 56.0 s for SVR. Multi-task Lasso and k-NN also fitted in approximately 0.03 s, whereas MLP and TabNet required 20.5 and 24.7 s, CatBoost 33.3 s, and SVR 56.0 s. SVR showed the steepest growth, from 0.054 s at $N = 500$ to 56.0 s at $N = 10{,}000$, consistent with the scaling of kernel fitting. CatBoost remained near 31–33 s because its selected boosting work and fixed overhead dominated the change in row count. These fit times exclude the separate 100-trial hyperparameter searches and saving of fitted surrogates; they therefore describe refitting cost once settings have been selected. Complete values are reported in Table S5.

The timing results sharpen the accuracy findings. MLP delivered the best full-size accuracy with moderate fitting time, while CatBoost and SVR were slower to refit under the selected settings. The multi-task linear surrogates were extremely fast and benefited from projection, but their final nRMSE remained higher than the nonlinear alternatives. Thus, projection does not remove the need to choose a surrogate appropriate to the desired accuracy–training-cost tradeoff; it adds a common feasibility layer after that choice has been made.

At $N = 10{,}000$, complete 20-component raw inference ranged from approximately 0.001 ms per sample for PLS and the multi-task linear surrogates to 1.874 ms per sample for SVR (Figure S2). Projection-only latency ranged from 0.068 to 0.144 ms per sample across surrogates. The projection therefore dominated end-to-end latency for the fastest linear predictors but remained below 0.15 ms per sample under the stated batching protocol. All 766,350 production projections used the QR solver; the SVD fallback was not activated. Consequently, projection supplied deterministic mass-conservation and non-negativity enforcement at a small absolute deployment cost. This makes the method especially attractive for downstream screening, optimization, or decision-support loops in which infeasible concentration vectors are more damaging than a sub-millisecond correction step.

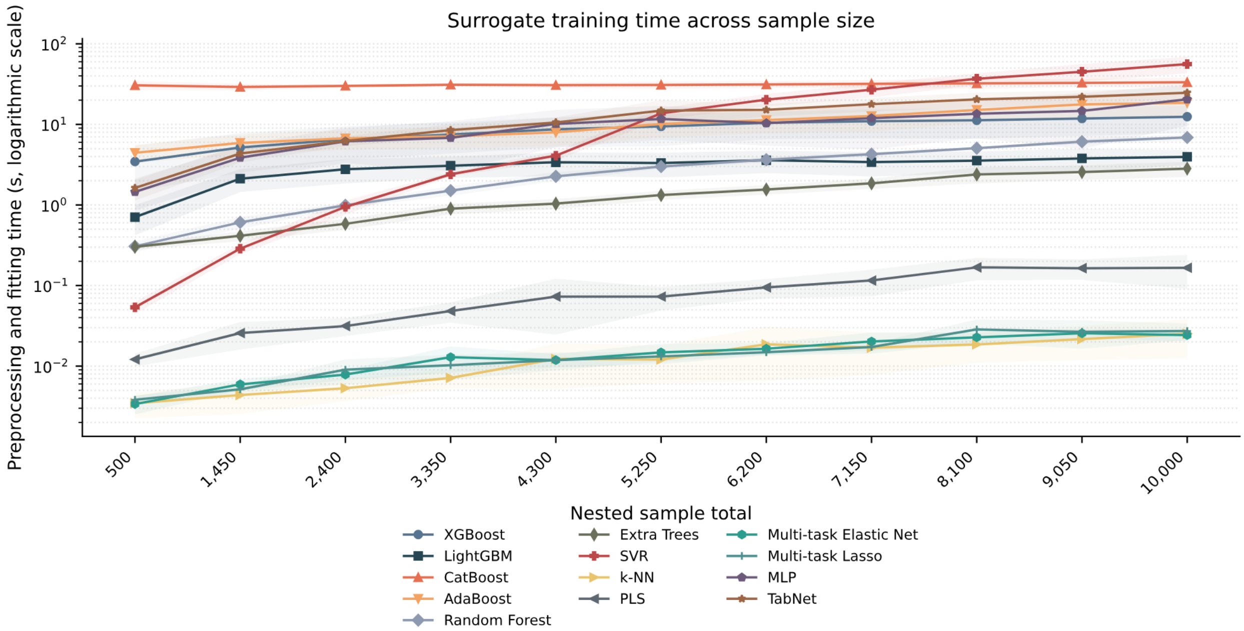

Figure S1: In-distribution preprocessing and surrogate-fitting time across nested sample totals. Lines show five-fold means, shaded regions show ± one sample SD, and the vertical axis is logarithmic. Hyperparameter-search time is excluded.

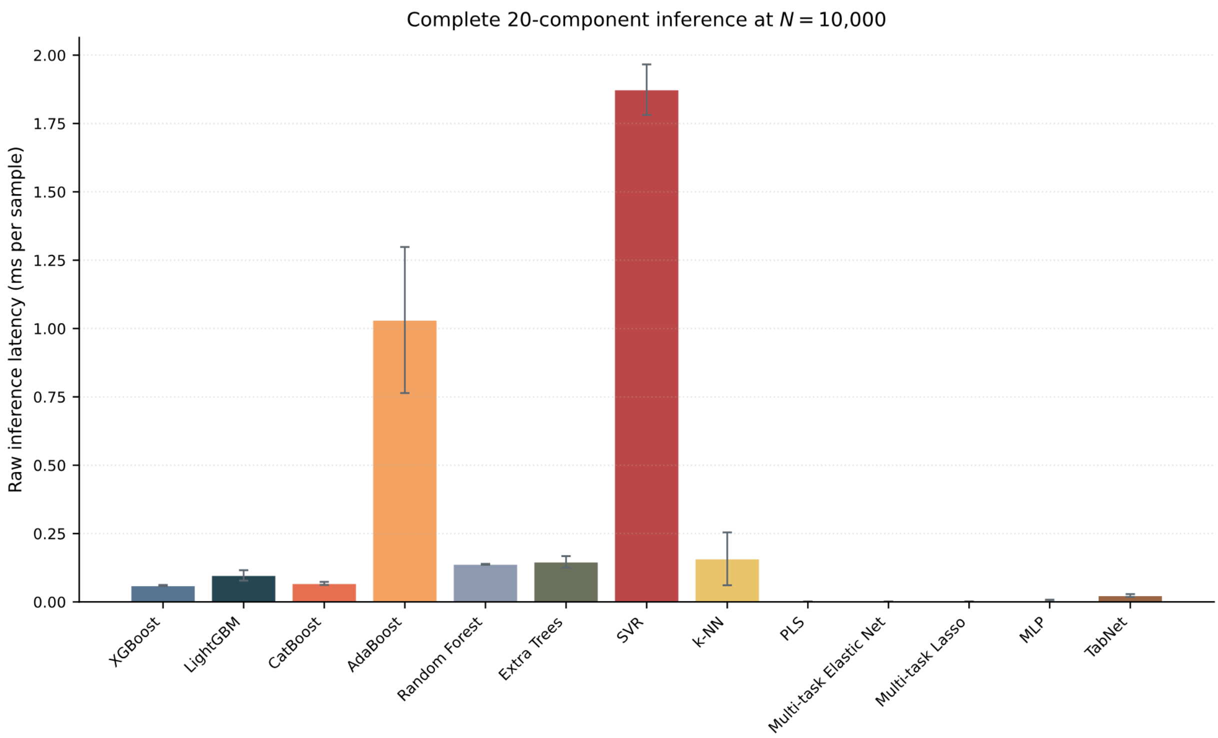

Figure S2: Complete 20-component raw surrogate-inference latency at $N = 10{,}000$. Bars show the five-fold mean and error bars show ± one sample SD of fold medians.

Table S5: Computational timing summary. Fit time includes fold-local preprocessing and surrogate fitting but excludes hyperparameter search; latency is milliseconds per sample. Values are five-fold mean ± sample SD.

Surrogate	Fit at $N = 500$ (s)	Fit at $N = 10{,}000$ (s)	Fit-time ratio	Raw latency	Projection latency
XGBoost	3.457 ± 2.000	12.458 ± 4.648	3.6	0.060 ± 0.002	0.098 ± 0.015
LightGBM	0.707 ± 0.285	3.957 ± 0.828	5.6	0.097 ± 0.019	0.084 ± 0.012
CatBoost	30.592 ± 3.762	33.303 ± 5.028	1.1	0.068 ± 0.006	0.123 ± 0.052
AdaBoost	4.439 ± 1.134	18.548 ± 6.043	4.2	1.031 ± 0.267	0.090 ± 0.041
Random Forest	0.307 ± 0.045	6.887 ± 0.643	22.5	0.138 ± 0.002	0.068 ± 0.004
Extra Trees	0.303 ± 0.023	2.832 ± 0.605	9.4	0.146 ± 0.021	0.105 ± 0.045
SVR	0.054 ± 0.009	56.021 ± 12.938	1045.3	1.874 ± 0.093	0.092 ± 0.042
k-NN	0.003 ± 0.001	0.025 ± 0.013	7.3	0.157 ± 0.097	0.140 ± 0.003
PLS	0.012 ± 0.002	0.166 ± 0.075	13.7	0.001 ± 0.000	0.144 ± 0.013
Multi-task Elastic Net	0.003 ± 0.001	0.024 ± 0.004	7.2	0.001 ± 0.001	0.141 ± 0.012
Multi-task Lasso	0.004 ± 0.001	0.027 ± 0.006	7.1	0.001 ± 0.001	0.139 ± 0.016
MLP	1.451 ± 0.590	20.459 ± 10.824	14.1	0.005 ± 0.003	0.115 ± 0.046
TabNet	1.629 ± 0.496	24.731 ± 4.059	15.2	0.023 ± 0.005	0.140 ± 0.041

6 Detailed Out-of-Distribution Results

The mild and severe out-of-distribution tables use the same 600 cases before and after projection. Component nRMSE uses the fixed full in-distribution component SD, and composite nRMSE uses the corresponding full in-distribution composite SD. Projection lowered aggregate 20-component nRMSE for 11 of 13 surrogates under mild extrapolation and 9 of 13 under severe extrapolation. At component level, it lowered 192 of 260 mild comparisons (73.8%) and 193 of 260 severe comparisons (74.2%).

6.1 Mild Extrapolation

Table S6: Mild out-of-distribution accuracy by effluent component and surrogate. Each value summarizes the same 600 mechanistic cases; Δ is projected minus raw.

Component	Unit	Surrogate	Raw nRMSE$_j$	Projected nRMSE$_j$	ΔnRMSE$_j$	Raw R_j^2	Projected R_j^2
S_O	g O_2 m^{-3}	XGBoost	0.389	0.389	−0.000	0.920	0.920
S_O	g O_2 m^{-3}	LightGBM	0.363	0.363	−0.000	0.930	0.930
S_O	g O_2 m^{-3}	CatBoost	0.339	0.339	−0.000	0.939	0.939
S_O	g O_2 m^{-3}	AdaBoost	0.730	0.731	+0.001	0.717	0.717
S_O	g O_2 m^{-3}	Random Forest	1.187	1.191	+0.004	0.253	0.248
S_O	g O_2 m^{-3}	Extra Trees	0.714	0.715	+0.001	0.730	0.729
S_O	g O_2 m^{-3}	SVR	0.267	0.268	+0.001	0.962	0.962
S_O	g O_2 m^{-3}	k-NN	0.827	0.827	+0.001	0.638	0.637
S_O	g O_2 m^{-3}	PLS	0.659	0.607	−0.052	0.770	0.805
S_O	g O_2 m^{-3}	Multi-task Elastic Net	0.590	0.504	−0.086	0.815	0.866
S_O	g O_2 m^{-3}	Multi-task Lasso	0.590	0.504	−0.087	0.815	0.866
S_O	g O_2 m^{-3}	MLP	0.155	0.154	−0.001	0.987	0.987
S_O	g O_2 m^{-3}	TabNet	0.360	0.356	−0.004	0.931	0.933
S_F	g COD m^{-3}	XGBoost	1.103	1.103	+0.000	0.624	0.624
S_F	g COD m^{-3}	LightGBM	1.096	1.096	+0.000	0.629	0.629
S_F	g COD m^{-3}	CatBoost	1.018	1.018	+0.000	0.679	0.679
S_F	g COD m^{-3}	AdaBoost	1.302	1.311	+0.009	0.476	0.469
S_F	g COD m^{-3}	Random Forest	1.679	2.271	+0.592	0.128	−0.595
S_F	g COD m^{-3}	Extra Trees	1.457	1.599	+0.142	0.343	0.209
S_F	g COD m^{-3}	SVR	1.652	1.655	+0.003	0.156	0.153
S_F	g COD m^{-3}	k-NN	1.669	1.674	+0.005	0.139	0.134
S_F	g COD m^{-3}	PLS	1.782	2.108	+0.326	0.018	−0.373
S_F	g COD m^{-3}	Multi-task Elastic Net	1.734	1.720	−0.014	0.071	0.085
S_F	g COD m^{-3}	Multi-task Lasso	1.734	1.720	−0.014	0.071	0.085
S_F	g COD m^{-3}	MLP	0.844	0.843	−0.002	0.780	0.780
S_F	g COD m^{-3}	TabNet	0.696	0.689	−0.007	0.850	0.853
S_A	g COD m^{-3}	XGBoost	0.262	0.262	−0.000	0.874	0.875
S_A	g COD m^{-3}	LightGBM	0.233	0.233	−0.000	0.901	0.901
S_A	g COD m^{-3}	CatBoost	0.231	0.228	−0.002	0.903	0.905
S_A	g COD m^{-3}	AdaBoost	0.531	0.531	+0.000	0.486	0.485
S_A	g COD m^{-3}	Random Forest	0.443	0.443	+0.001	0.643	0.642
S_A	g COD m^{-3}	Extra Trees	0.395	0.394	−0.000	0.716	0.716
S_A	g COD m^{-3}	SVR	0.435	0.393	−0.042	0.655	0.719
S_A	g COD m^{-3}	k-NN	0.552	0.553	+0.000	0.443	0.443
S_A	g COD m^{-3}	PLS	0.789	0.586	−0.203	−0.135	0.373
S_A	g COD m^{-3}	Multi-task Elastic Net	0.776	0.549	−0.226	−0.098	0.450

Continued on next page

Table S6 continued

Component	Unit	Surrogate	Raw nRMSE$_j$	Projected nRMSE$_j$	ΔnRMSE$_j$	Raw R_j^2	Projected R_j^2
S_A	g COD m^{-3}	Multi-task Lasso	0.776	0.549	−0.227	−0.098	0.450
S_A	g COD m^{-3}	MLP	0.136	0.128	−0.008	0.966	0.970
S_A	g COD m^{-3}	TabNet	0.344	0.343	−0.000	0.785	0.785
S_{NH4}	g N m^{-3}	XGBoost	0.372	0.273	−0.099	0.915	0.954
S_{NH4}	g N m^{-3}	LightGBM	0.357	0.270	−0.087	0.922	0.955
S_{NH4}	g N m^{-3}	CatBoost	0.351	0.256	−0.096	0.924	0.960
S_{NH4}	g N m^{-3}	AdaBoost	0.667	0.598	−0.069	0.726	0.780
S_{NH4}	g N m^{-3}	Random Forest	1.013	0.728	−0.285	0.370	0.674
S_{NH4}	g N m^{-3}	Extra Trees	0.737	0.458	−0.279	0.667	0.871
S_{NH4}	g N m^{-3}	SVR	0.386	0.358	−0.028	0.909	0.921
S_{NH4}	g N m^{-3}	k-NN	0.772	0.578	−0.194	0.634	0.795
S_{NH4}	g N m^{-3}	PLS	0.937	0.619	−0.319	0.461	0.765
S_{NH4}	g N m^{-3}	Multi-task Elastic Net	0.814	0.583	−0.231	0.593	0.791
S_{NH4}	g N m^{-3}	Multi-task Lasso	0.814	0.583	−0.231	0.593	0.791
S_{NH4}	g N m^{-3}	MLP	0.180	0.136	−0.044	0.980	0.989
S_{NH4}	g N m^{-3}	TabNet	0.192	0.164	−0.028	0.977	0.983
S_{NO2}	g N m^{-3}	XGBoost	0.765	0.731	−0.033	0.644	0.674
S_{NO2}	g N m^{-3}	LightGBM	0.771	0.736	−0.035	0.638	0.670
S_{NO2}	g N m^{-3}	CatBoost	0.736	0.696	−0.040	0.670	0.705
S_{NO2}	g N m^{-3}	AdaBoost	1.202	1.183	−0.019	0.120	0.148
S_{NO2}	g N m^{-3}	Random Forest	1.221	1.181	−0.040	0.092	0.150
S_{NO2}	g N m^{-3}	Extra Trees	0.891	0.829	−0.061	0.517	0.581
S_{NO2}	g N m^{-3}	SVR	0.963	0.800	−0.163	0.435	0.610
S_{NO2}	g N m^{-3}	k-NN	1.048	1.011	−0.038	0.331	0.378
S_{NO2}	g N m^{-3}	PLS	1.176	1.146	−0.031	0.157	0.201
S_{NO2}	g N m^{-3}	Multi-task Elastic Net	1.156	1.147	−0.009	0.186	0.199
S_{NO2}	g N m^{-3}	Multi-task Lasso	1.156	1.147	−0.009	0.186	0.199
S_{NO2}	g N m^{-3}	MLP	0.254	0.240	−0.014	0.961	0.965
S_{NO2}	g N m^{-3}	TabNet	0.471	0.484	+0.014	0.865	0.857
S_{NO3}	g N m^{-3}	XGBoost	0.461	0.429	−0.032	0.892	0.907
S_{NO3}	g N m^{-3}	LightGBM	0.454	0.434	−0.021	0.895	0.905
S_{NO3}	g N m^{-3}	CatBoost	0.428	0.399	−0.028	0.907	0.919
S_{NO3}	g N m^{-3}	AdaBoost	0.822	0.818	−0.004	0.657	0.660
S_{NO3}	g N m^{-3}	Random Forest	1.174	0.987	−0.187	0.301	0.506
S_{NO3}	g N m^{-3}	Extra Trees	0.684	0.619	−0.065	0.763	0.805
S_{NO3}	g N m^{-3}	SVR	0.464	0.429	−0.035	0.891	0.906
S_{NO3}	g N m^{-3}	k-NN	0.859	0.695	−0.164	0.625	0.755
S_{NO3}	g N m^{-3}	PLS	0.973	0.623	−0.350	0.519	0.803
S_{NO3}	g N m^{-3}	Multi-task Elastic Net	0.869	0.722	−0.147	0.617	0.735
S_{NO3}	g N m^{-3}	Multi-task Lasso	0.869	0.722	−0.147	0.617	0.736
S_{NO3}	g N m^{-3}	MLP	0.195	0.150	−0.045	0.981	0.989
S_{NO3}	g N m^{-3}	TabNet	0.330	0.218	−0.112	0.945	0.976
S_{N2}	g N m^{-3}	XGBoost	0.599	0.558	−0.041	0.758	0.790
S_{N2}	g N m^{-3}	LightGBM	0.608	0.548	−0.060	0.750	0.797
S_{N2}	g N m^{-3}	CatBoost	0.546	0.517	−0.029	0.799	0.820
S_{N2}	g N m^{-3}	AdaBoost	0.869	0.832	−0.037	0.490	0.532
S_{N2}	g N m^{-3}	Random Forest	1.047	0.942	−0.104	0.260	0.400
S_{N2}	g N m^{-3}	Extra Trees	0.750	0.678	−0.072	0.620	0.689
S_{N2}	g N m^{-3}	SVR	0.704	0.635	−0.069	0.665	0.728
S_{N2}	g N m^{-3}	k-NN	0.813	0.737	−0.076	0.553	0.633
S_{N2}	g N m^{-3}	PLS	0.965	0.893	−0.072	0.371	0.461
S_{N2}	g N m^{-3}	Multi-task Elastic Net	0.871	0.819	−0.052	0.487	0.547
S_{N2}	g N m^{-3}	Multi-task Lasso	0.871	0.819	−0.052	0.487	0.547
S_{N2}	g N m^{-3}	MLP	0.223	0.233	+0.011	0.967	0.963
S_{N2}	g N m^{-3}	TabNet	0.339	0.329	−0.010	0.922	0.927
S_{PO4}	g P m^{-3}	XGBoost	0.308	0.277	−0.031	0.897	0.917
S_{PO4}	g P m^{-3}	LightGBM	0.301	0.274	−0.027	0.902	0.919
S_{PO4}	g P m^{-3}	CatBoost	0.271	0.256	−0.016	0.920	0.929
S_{PO4}	g P m^{-3}	AdaBoost	0.607	0.539	−0.068	0.602	0.686
S_{PO4}	g P m^{-3}	Random Forest	0.835	0.550	−0.284	0.247	0.672
S_{PO4}	g P m^{-3}	Extra Trees	0.654	0.432	−0.223	0.537	0.798
S_{PO4}	g P m^{-3}	SVR	0.380	0.374	−0.006	0.844	0.849
S_{PO4}	g P m^{-3}	k-NN	0.566	0.471	−0.095	0.654	0.760
S_{PO4}	g P m^{-3}	PLS	0.708	0.565	−0.143	0.458	0.655
S_{PO4}	g P m^{-3}	Multi-task Elastic Net	0.606	0.544	−0.062	0.603	0.680
S_{PO4}	g P m^{-3}	Multi-task Lasso	0.606	0.544	−0.062	0.603	0.680
S_{PO4}	g P m^{-3}	MLP	0.179	0.157	−0.022	0.965	0.973
S_{PO4}	g P m^{-3}	TabNet	0.353	0.308	−0.045	0.865	0.898
S_I	g COD m^{-3}	XGBoost	0.017	0.000	−0.017	1.000	1.000
S_I	g COD m^{-3}	LightGBM	0.004	0.000	−0.004	1.000	1.000
S_I	g COD m^{-3}	CatBoost	0.006	0.000	−0.006	1.000	1.000
S_I	g COD m^{-3}	AdaBoost	0.088	0.000	−0.088	0.992	1.000
S_I	g COD m^{-3}	Random Forest	0.433	0.000	−0.433	0.813	1.000
S_I	g COD m^{-3}	Extra Trees	0.478	0.000	−0.478	0.772	1.000
S_I	g COD m^{-3}	SVR	0.121	0.000	−0.121	0.985	1.000
S_I	g COD m^{-3}	k-NN	0.496	0.000	−0.496	0.754	1.000
S_I	g COD m^{-3}	PLS	0.174	0.000	−0.174	0.970	1.000
S_I	g COD m^{-3}	Multi-task Elastic Net	0.002	0.000	−0.002	1.000	1.000
S_I	g COD m^{-3}	Multi-task Lasso	0.002	0.000	−0.002	1.000	1.000
S_I	g COD m^{-3}	MLP	0.096	0.000	−0.096	0.991	1.000

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Table S6 continued

Component	Unit	Surrogate	Raw nRMSE$_j$	Projected nRMSE$_j$	ΔnRMSE$_j$	Raw R_j^2	Projected R_j^2
S_I	g COD m^{-3}	TabNet	0.163	0.000	−0.163	0.973	1.000
S_{ALK}	mol m^{-3}	XGBoost	0.020	0.010	−0.011	1.000	1.000
S_{ALK}	mol m^{-3}	LightGBM	0.017	0.009	−0.008	1.000	1.000
S_{ALK}	mol m^{-3}	CatBoost	0.015	0.009	−0.006	1.000	1.000
S_{ALK}	mol m^{-3}	AdaBoost	0.089	0.021	−0.068	0.992	1.000
S_{ALK}	mol m^{-3}	Random Forest	0.194	0.025	−0.169	0.962	0.999
S_{ALK}	mol m^{-3}	Extra Trees	0.476	0.015	−0.461	0.773	1.000
S_{ALK}	mol m^{-3}	SVR	0.123	0.012	−0.111	0.985	1.000
S_{ALK}	mol m^{-3}	k-NN	0.500	0.020	−0.480	0.749	1.000
S_{ALK}	mol m^{-3}	PLS	0.456	0.018	−0.438	0.792	1.000
S_{ALK}	mol m^{-3}	Multi-task Elastic Net	0.028	0.019	−0.008	0.999	1.000
S_{ALK}	mol m^{-3}	Multi-task Lasso	0.028	0.019	−0.008	0.999	1.000
S_{ALK}	mol m^{-3}	MLP	0.089	0.004	−0.085	0.992	1.000
S_{ALK}	mol m^{-3}	TabNet	0.172	0.005	−0.167	0.970	1.000
X_I	g COD m^{-3}	XGBoost	0.064	0.079	+0.015	0.996	0.994
X_I	g COD m^{-3}	LightGBM	0.057	0.069	+0.012	0.997	0.995
X_I	g COD m^{-3}	CatBoost	0.050	0.061	+0.010	0.997	0.996
X_I	g COD m^{-3}	AdaBoost	0.201	0.244	+0.044	0.959	0.940
X_I	g COD m^{-3}	Random Forest	0.355	0.375	+0.020	0.873	0.858
X_I	g COD m^{-3}	Extra Trees	0.494	0.490	−0.004	0.754	0.758
X_I	g COD m^{-3}	SVR	0.132	0.148	+0.016	0.983	0.978
X_I	g COD m^{-3}	k-NN	0.490	0.489	−0.001	0.758	0.759
X_I	g COD m^{-3}	PLS	0.556	0.563	+0.006	0.688	0.681
X_I	g COD m^{-3}	Multi-task Elastic Net	0.048	0.069	+0.021	0.998	0.995
X_I	g COD m^{-3}	Multi-task Lasso	0.048	0.069	+0.021	0.998	0.995
X_I	g COD m^{-3}	MLP	0.087	0.093	+0.006	0.992	0.991
X_I	g COD m^{-3}	TabNet	0.222	0.226	+0.004	0.951	0.949
X_S	g COD m^{-3}	XGBoost	0.373	0.373	−0.000	0.918	0.918
X_S	g COD m^{-3}	LightGBM	0.346	0.346	−0.000	0.930	0.930
X_S	g COD m^{-3}	CatBoost	0.337	0.337	+0.000	0.933	0.933
X_S	g COD m^{-3}	AdaBoost	0.641	0.647	+0.006	0.759	0.755
X_S	g COD m^{-3}	Random Forest	0.620	0.668	+0.048	0.775	0.738
X_S	g COD m^{-3}	Extra Trees	0.678	0.688	+0.009	0.731	0.723
X_S	g COD m^{-3}	SVR	0.383	0.386	+0.004	0.914	0.913
X_S	g COD m^{-3}	k-NN	0.869	0.873	+0.003	0.557	0.554
X_S	g COD m^{-3}	PLS	1.086	0.976	−0.110	0.309	0.442
X_S	g COD m^{-3}	Multi-task Elastic Net	0.921	0.711	−0.210	0.503	0.704
X_S	g COD m^{-3}	Multi-task Lasso	0.921	0.711	−0.210	0.503	0.704
X_S	g COD m^{-3}	MLP	0.120	0.120	−0.000	0.992	0.992
X_S	g COD m^{-3}	TabNet	0.213	0.212	−0.001	0.973	0.974
X_H	g COD m^{-3}	XGBoost	0.257	0.450	+0.193	0.948	0.840
X_H	g COD m^{-3}	LightGBM	0.254	0.420	+0.166	0.949	0.861
X_H	g COD m^{-3}	CatBoost	0.231	0.382	+0.152	0.958	0.885
X_H	g COD m^{-3}	AdaBoost	0.706	1.187	+0.482	0.608	−0.111
X_H	g COD m^{-3}	Random Forest	0.521	1.963	+1.442	0.786	−2.037
X_H	g COD m^{-3}	Extra Trees	0.857	1.124	+0.267	0.421	0.004
X_H	g COD m^{-3}	SVR	0.279	0.498	+0.219	0.939	0.804
X_H	g COD m^{-3}	k-NN	0.655	1.069	+0.415	0.662	0.099
X_H	g COD m^{-3}	PLS	1.035	0.986	−0.048	0.157	0.234
X_H	g COD m^{-3}	Multi-task Elastic Net	0.619	0.641	+0.022	0.698	0.677
X_H	g COD m^{-3}	Multi-task Lasso	0.619	0.641	+0.022	0.698	0.677
X_H	g COD m^{-3}	MLP	0.148	0.273	+0.124	0.983	0.941
X_H	g COD m^{-3}	TabNet	0.245	0.531	+0.287	0.953	0.778
X_{PAO}	g COD m^{-3}	XGBoost	0.142	0.148	+0.006	0.979	0.977
X_{PAO}	g COD m^{-3}	LightGBM	0.145	0.150	+0.005	0.978	0.977
X_{PAO}	g COD m^{-3}	CatBoost	0.124	0.130	+0.006	0.984	0.983
X_{PAO}	g COD m^{-3}	AdaBoost	0.251	0.261	+0.010	0.934	0.929
X_{PAO}	g COD m^{-3}	Random Forest	0.583	0.575	−0.008	0.647	0.656
X_{PAO}	g COD m^{-3}	Extra Trees	0.391	0.387	−0.004	0.841	0.844
X_{PAO}	g COD m^{-3}	SVR	0.201	0.205	+0.004	0.958	0.956
X_{PAO}	g COD m^{-3}	k-NN	0.482	0.475	−0.006	0.759	0.765
X_{PAO}	g COD m^{-3}	PLS	0.644	0.654	+0.011	0.569	0.554
X_{PAO}	g COD m^{-3}	Multi-task Elastic Net	0.236	0.219	−0.017	0.942	0.950
X_{PAO}	g COD m^{-3}	Multi-task Lasso	0.236	0.219	−0.017	0.942	0.950
X_{PAO}	g COD m^{-3}	MLP	0.134	0.137	+0.003	0.981	0.980
X_{PAO}	g COD m^{-3}	TabNet	0.253	0.260	+0.007	0.933	0.930
X_{PP}	g P m^{-3}	XGBoost	0.331	0.351	+0.020	0.870	0.853
X_{PP}	g P m^{-3}	LightGBM	0.335	0.349	+0.014	0.867	0.855
X_{PP}	g P m^{-3}	CatBoost	0.320	0.323	+0.003	0.878	0.876
X_{PP}	g P m^{-3}	AdaBoost	0.632	0.648	+0.016	0.525	0.501
X_{PP}	g P m^{-3}	Random Forest	0.725	0.731	+0.006	0.375	0.365
X_{PP}	g P m^{-3}	Extra Trees	0.604	0.555	−0.050	0.566	0.634
X_{PP}	g P m^{-3}	SVR	0.454	0.448	−0.006	0.755	0.762
X_{PP}	g P m^{-3}	k-NN	0.585	0.590	+0.005	0.593	0.586
X_{PP}	g P m^{-3}	PLS	0.848	0.725	−0.123	0.145	0.375
X_{PP}	g P m^{-3}	Multi-task Elastic Net	0.684	0.671	−0.013	0.444	0.465
X_{PP}	g P m^{-3}	Multi-task Lasso	0.684	0.671	−0.013	0.444	0.465
X_{PP}	g P m^{-3}	MLP	0.211	0.197	−0.015	0.947	0.954
X_{PP}	g P m^{-3}	TabNet	0.358	0.363	+0.006	0.848	0.843
X_{PHA}	g COD m^{-3}	XGBoost	0.374	0.373	−0.001	0.846	0.846

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Table S6 continued

Component	Unit	Surrogate	Raw nRMSE_j	Projected nRMSE_j	$\Delta\mathrm{nRMSE}_j$	Raw R_j^2	Projected R_j^2
X_{PHA}	g COD m^{-3}	LightGBM	0.366	0.365	−0.001	0.852	0.853
X_{PHA}	g COD m^{-3}	CatBoost	0.356	0.354	−0.002	0.861	0.862
X_{PHA}	g COD m^{-3}	AdaBoost	0.723	0.723	−0.000	0.424	0.424
X_{PHA}	g COD m^{-3}	Random Forest	0.756	0.753	−0.002	0.371	0.375
X_{PHA}	g COD m^{-3}	Extra Trees	0.620	0.619	−0.001	0.576	0.578
X_{PHA}	g COD m^{-3}	SVR	0.526	0.520	−0.006	0.696	0.702
X_{PHA}	g COD m^{-3}	k-NN	0.628	0.628	+0.000	0.566	0.566
X_{PHA}	g COD m^{-3}	PLS	0.865	0.770	−0.095	0.176	0.348
X_{PHA}	g COD m^{-3}	Multi-task Elastic Net	0.764	0.619	−0.145	0.357	0.578
X_{PHA}	g COD m^{-3}	Multi-task Lasso	0.764	0.619	−0.145	0.356	0.578
X_{PHA}	g COD m^{-3}	MLP	0.221	0.203	−0.018	0.946	0.955
X_{PHA}	g COD m^{-3}	TabNet	0.399	0.375	−0.024	0.825	0.845
X_{AOB}	g COD m^{-3}	XGBoost	0.244	0.241	−0.003	0.953	0.954
X_{AOB}	g COD m^{-3}	LightGBM	0.243	0.240	−0.003	0.954	0.955
X_{AOB}	g COD m^{-3}	CatBoost	0.220	0.216	−0.004	0.962	0.963
X_{AOB}	g COD m^{-3}	AdaBoost	0.661	0.661	+0.000	0.657	0.657
X_{AOB}	g COD m^{-3}	Random Forest	1.020	1.006	−0.014	0.183	0.206
X_{AOB}	g COD m^{-3}	Extra Trees	0.631	0.631	−0.000	0.687	0.687
X_{AOB}	g COD m^{-3}	SVR	0.284	0.289	+0.005	0.937	0.934
X_{AOB}	g COD m^{-3}	k-NN	0.615	0.611	−0.005	0.703	0.707
X_{AOB}	g COD m^{-3}	PLS	0.763	0.764	+0.001	0.543	0.542
X_{AOB}	g COD m^{-3}	Multi-task Elastic Net	0.522	0.515	−0.007	0.786	0.792
X_{AOB}	g COD m^{-3}	Multi-task Lasso	0.522	0.515	−0.007	0.786	0.792
X_{AOB}	g COD m^{-3}	MLP	0.130	0.129	−0.001	0.987	0.987
X_{AOB}	g COD m^{-3}	TabNet	0.208	0.207	−0.001	0.966	0.966
X_{NOB}	g COD m^{-3}	XGBoost	0.215	0.215	−0.001	0.959	0.959
X_{NOB}	g COD m^{-3}	LightGBM	0.211	0.210	−0.001	0.960	0.961
X_{NOB}	g COD m^{-3}	CatBoost	0.189	0.188	−0.001	0.968	0.968
X_{NOB}	g COD m^{-3}	AdaBoost	0.428	0.426	−0.002	0.836	0.838
X_{NOB}	g COD m^{-3}	Random Forest	0.982	0.975	−0.007	0.140	0.152
X_{NOB}	g COD m^{-3}	Extra Trees	0.520	0.519	−0.001	0.759	0.760
X_{NOB}	g COD m^{-3}	SVR	0.292	0.294	+0.002	0.924	0.923
X_{NOB}	g COD m^{-3}	k-NN	0.546	0.544	−0.003	0.734	0.736
X_{NOB}	g COD m^{-3}	PLS	0.931	0.922	−0.009	0.226	0.241
X_{NOB}	g COD m^{-3}	Multi-task Elastic Net	0.380	0.374	−0.006	0.871	0.875
X_{NOB}	g COD m^{-3}	Multi-task Lasso	0.380	0.374	−0.006	0.871	0.875
X_{NOB}	g COD m^{-3}	MLP	0.127	0.127	−0.000	0.986	0.986
X_{NOB}	g COD m^{-3}	TabNet	0.217	0.216	−0.000	0.958	0.958
X_{MeP}	g TSS m^{-3}	XGBoost	0.242	0.217	−0.025	0.942	0.954
X_{MeP}	g TSS m^{-3}	LightGBM	0.224	0.205	−0.018	0.950	0.958
X_{MeP}	g TSS m^{-3}	CatBoost	0.217	0.208	−0.009	0.953	0.957
X_{MeP}	g TSS m^{-3}	AdaBoost	0.554	0.416	−0.138	0.696	0.828
X_{MeP}	g TSS m^{-3}	Random Forest	0.940	0.543	−0.397	0.124	0.707
X_{MeP}	g TSS m^{-3}	Extra Trees	0.784	0.396	−0.389	0.390	0.845
X_{MeP}	g TSS m^{-3}	SVR	0.284	0.246	−0.038	0.920	0.940
X_{MeP}	g TSS m^{-3}	k-NN	0.553	0.376	−0.177	0.697	0.860
X_{MeP}	g TSS m^{-3}	PLS	0.824	0.392	−0.432	0.327	0.848
X_{MeP}	g TSS m^{-3}	Multi-task Elastic Net	0.419	0.400	−0.019	0.826	0.841
X_{MeP}	g TSS m^{-3}	Multi-task Lasso	0.419	0.400	−0.019	0.826	0.841
X_{MeP}	g TSS m^{-3}	MLP	0.184	0.167	−0.017	0.967	0.972
X_{MeP}	g TSS m^{-3}	TabNet	0.313	0.218	−0.094	0.903	0.953
X_{MeOH}	g TSS m^{-3}	XGBoost	0.365	0.362	−0.003	0.893	0.895
X_{MeOH}	g TSS m^{-3}	LightGBM	0.344	0.344	−0.000	0.905	0.905
X_{MeOH}	g TSS m^{-3}	CatBoost	0.352	0.348	−0.005	0.901	0.903
X_{MeOH}	g TSS m^{-3}	AdaBoost	0.709	0.696	−0.013	0.597	0.612
X_{MeOH}	g TSS m^{-3}	Random Forest	1.021	0.909	−0.112	0.165	0.337
X_{MeOH}	g TSS m^{-3}	Extra Trees	0.783	0.662	−0.121	0.509	0.649
X_{MeOH}	g TSS m^{-3}	SVR	0.444	0.412	−0.032	0.842	0.864
X_{MeOH}	g TSS m^{-3}	k-NN	0.709	0.628	−0.080	0.597	0.684
X_{MeOH}	g TSS m^{-3}	PLS	0.902	0.656	−0.246	0.348	0.655
X_{MeOH}	g TSS m^{-3}	Multi-task Elastic Net	0.702	0.670	−0.032	0.606	0.641
X_{MeOH}	g TSS m^{-3}	Multi-task Lasso	0.702	0.670	−0.032	0.606	0.641
X_{MeOH}	g TSS m^{-3}	MLP	0.290	0.279	−0.011	0.933	0.938
X_{MeOH}	g TSS m^{-3}	TabNet	0.417	0.366	−0.051	0.861	0.893

Table S7: Mild out-of-distribution accuracy by derived water-quality composite and surrogate. Composite nRMSE is physical-unit RMSE divided by the population SD of that composite in all 10,000 in-distribution targets. Each value summarizes the same 600 mechanistic cases; Δ is projected minus raw.

Composite	Surrogate	Raw nRMSE$_q$	Projected nRMSE$_q$	ΔnRMSE$_q$	Raw R^2	Projected R^2
COD	XGBoost	0.283	0.382	+0.098	0.942	0.895
COD	LightGBM	0.263	0.346	+0.082	0.950	0.914
COD	CatBoost	0.255	0.332	+0.078	0.953	0.920
COD	AdaBoost	0.536	0.792	+0.256	0.793	0.547
COD	Random Forest	0.570	1.289	+0.719	0.765	−0.201
COD	Extra Trees	0.760	0.876	+0.116	0.582	0.445
COD	SVR	0.195	0.337	+0.142	0.972	0.918
COD	k-NN	0.634	0.832	+0.199	0.710	0.499
COD	PLS	0.778	0.839	+0.062	0.563	0.491
COD	Multi-task Elastic Net	0.298	0.450	+0.153	0.936	0.853
COD	Multi-task Lasso	0.298	0.450	+0.153	0.936	0.853
COD	MLP	0.104	0.167	+0.062	0.992	0.980
COD	TabNet	0.200	0.343	+0.143	0.971	0.915
TN	XGBoost	0.473	0.271	−0.203	0.869	0.957
TN	LightGBM	0.464	0.266	−0.198	0.874	0.959
TN	CatBoost	0.460	0.251	−0.209	0.876	0.963
TN	AdaBoost	0.699	0.404	−0.295	0.714	0.905
TN	Random Forest	1.397	0.458	−0.940	−0.141	0.878
TN	Extra Trees	0.901	0.329	−0.571	0.526	0.937
TN	SVR	0.417	0.308	−0.109	0.898	0.944
TN	k-NN	0.862	0.358	−0.504	0.566	0.925
TN	PLS	1.180	0.434	−0.746	0.187	0.890
TN	Multi-task Elastic Net	0.423	0.398	−0.026	0.895	0.908
TN	Multi-task Lasso	0.423	0.398	−0.026	0.895	0.908
TN	MLP	0.187	0.113	−0.073	0.980	0.993
TN	TabNet	0.367	0.160	−0.208	0.921	0.985
TP	XGBoost	0.137	0.000	−0.137	0.982	1.000
TP	LightGBM	0.122	0.000	−0.122	0.985	1.000
TP	CatBoost	0.096	0.000	−0.096	0.991	1.000
TP	AdaBoost	0.414	0.000	−0.414	0.833	1.000
TP	Random Forest	0.960	0.000	−0.960	0.102	1.000
TP	Extra Trees	0.748	0.000	−0.748	0.455	1.000
TP	SVR	0.130	0.000	−0.130	0.984	1.000
TP	k-NN	0.497	0.000	−0.496	0.760	1.000
TP	PLS	0.714	0.000	−0.714	0.504	1.000
TP	Multi-task Elastic Net	0.003	0.000	−0.003	1.000	1.000
TP	Multi-task Lasso	0.003	0.000	−0.003	1.000	1.000
TP	MLP	0.085	0.000	−0.085	0.993	1.000
TP	TabNet	0.236	0.000	−0.236	0.946	1.000
TSS	XGBoost	0.301	0.425	+0.124	0.937	0.875
TSS	LightGBM	0.285	0.381	+0.096	0.944	0.900
TSS	CatBoost	0.276	0.365	+0.089	0.947	0.908
TSS	AdaBoost	0.577	0.958	+0.381	0.770	0.366
TSS	Random Forest	0.642	1.532	+0.891	0.715	−0.622
TSS	Extra Trees	0.869	1.018	+0.149	0.479	0.285
TSS	SVR	0.229	0.406	+0.176	0.964	0.886
TSS	k-NN	0.670	0.925	+0.255	0.690	0.409
TSS	PLS	1.039	0.951	−0.088	0.254	0.375
TSS	Multi-task Elastic Net	0.284	0.504	+0.220	0.944	0.824
TSS	Multi-task Lasso	0.284	0.504	+0.220	0.944	0.824
TSS	MLP	0.131	0.212	+0.081	0.988	0.969
TSS	TabNet	0.269	0.471	+0.202	0.950	0.847

6.2 Severe Extrapolation

Table S8: Severe out-of-distribution accuracy by effluent component and surrogate. Each value summarizes the same 600 mechanistic cases; Δ is projected minus raw.

Component	Unit	Surrogate	Raw nRMSE$_j$	Projected nRMSE$_j$	ΔnRMSE$_j$	Raw R_j^2	Projected R_j^2
S_O	g O_2 m^{-3}	XGBoost	0.717	0.717	+0.000	0.784	0.784
S_O	g O_2 m^{-3}	LightGBM	0.690	0.690	−0.000	0.799	0.799
S_O	g O_2 m^{-3}	CatBoost	0.666	0.666	−0.000	0.813	0.813
S_O	g O_2 m^{-3}	AdaBoost	0.999	1.008	+0.009	0.579	0.572
S_O	g O_2 m^{-3}	Random Forest	1.408	1.410	+0.002	0.164	0.162
S_O	g O_2 m^{-3}	Extra Trees	1.007	1.018	+0.011	0.572	0.563
S_O	g O_2 m^{-3}	SVR	0.373	0.373	−0.000	0.941	0.941
S_O	g O_2 m^{-3}	k-NN	1.004	1.015	+0.011	0.575	0.566
S_O	g O_2 m^{-3}	PLS	0.695	0.619	−0.077	0.796	0.839
S_O	g O_2 m^{-3}	Multi-task Elastic Net	0.676	0.496	−0.180	0.807	0.896
S_O	g O_2 m^{-3}	Multi-task Lasso	0.676	0.496	−0.180	0.807	0.896
S_O	g O_2 m^{-3}	MLP	0.276	0.275	−0.001	0.968	0.968
S_O	g O_2 m^{-3}	TabNet	0.464	0.455	−0.008	0.909	0.913
S_F	g COD m^{-3}	XGBoost	2.951	2.951	−0.000	0.312	0.312
S_F	g COD m^{-3}	LightGBM	2.888	2.888	+0.000	0.342	0.342
S_F	g COD m^{-3}	CatBoost	2.865	2.865	−0.000	0.352	0.352
S_F	g COD m^{-3}	AdaBoost	3.050	3.689	+0.639	0.266	−0.074
S_F	g COD m^{-3}	Random Forest	3.517	4.755	+1.238	0.024	−0.785

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Table S8 continued

Component	Unit	Surrogate	Raw nRMSE$_j$	Projected nRMSE$_j$	ΔnRMSE$_j$	Raw R_j^2	Projected R_j^2
S_F	g COD m^{-3}	Extra Trees	3.225	4.247	+1.022	0.179	−0.424
S_F	g COD m^{-3}	SVR	3.431	3.426	−0.005	0.071	0.074
S_F	g COD m^{-3}	k-NN	3.461	3.841	+0.379	0.054	−0.164
S_F	g COD m^{-3}	PLS	3.549	3.888	+0.339	0.006	−0.193
S_F	g COD m^{-3}	Multi-task Elastic Net	3.506	3.485	−0.020	0.030	0.041
S_F	g COD m^{-3}	Multi-task Lasso	3.506	3.485	−0.020	0.030	0.041
S_F	g COD m^{-3}	MLP	1.967	1.965	−0.002	0.695	0.695
S_F	g COD m^{-3}	TabNet	1.683	1.677	−0.006	0.776	0.778
S_A	g COD m^{-3}	XGBoost	0.288	0.288	−0.000	0.861	0.862
S_A	g COD m^{-3}	LightGBM	0.294	0.294	−0.000	0.855	0.855
S_A	g COD m^{-3}	CatBoost	0.284	0.282	−0.001	0.866	0.867
S_A	g COD m^{-3}	AdaBoost	0.545	0.569	+0.024	0.504	0.460
S_A	g COD m^{-3}	Random Forest	0.493	0.551	+0.058	0.594	0.494
S_A	g COD m^{-3}	Extra Trees	0.424	0.476	+0.051	0.699	0.622
S_A	g COD m^{-3}	SVR	0.554	0.409	−0.145	0.487	0.720
S_A	g COD m^{-3}	k-NN	0.581	0.595	+0.014	0.436	0.408
S_A	g COD m^{-3}	PLS	0.971	0.616	−0.355	−0.576	0.365
S_A	g COD m^{-3}	Multi-task Elastic Net	0.955	0.545	−0.410	−0.524	0.503
S_A	g COD m^{-3}	Multi-task Lasso	0.955	0.545	−0.410	−0.525	0.503
S_A	g COD m^{-3}	MLP	0.197	0.177	−0.020	0.935	0.947
S_A	g COD m^{-3}	TabNet	0.449	0.448	−0.000	0.664	0.664
S_{NH4}	g N m^{-3}	XGBoost	0.827	0.471	−0.357	0.760	0.922
S_{NH4}	g N m^{-3}	LightGBM	0.809	0.478	−0.330	0.771	0.920
S_{NH4}	g N m^{-3}	CatBoost	0.797	0.449	−0.348	0.777	0.929
S_{NH4}	g N m^{-3}	AdaBoost	1.074	0.731	−0.343	0.596	0.813
S_{NH4}	g N m^{-3}	Random Forest	1.445	0.953	−0.492	0.267	0.682
S_{NH4}	g N m^{-3}	Extra Trees	1.209	0.649	−0.560	0.487	0.852
S_{NH4}	g N m^{-3}	SVR	0.748	0.536	−0.212	0.804	0.899
S_{NH4}	g N m^{-3}	k-NN	1.163	0.727	−0.437	0.525	0.815
S_{NH4}	g N m^{-3}	PLS	1.410	0.794	−0.616	0.303	0.779
S_{NH4}	g N m^{-3}	Multi-task Elastic Net	1.157	0.791	−0.366	0.530	0.780
S_{NH4}	g N m^{-3}	Multi-task Lasso	1.157	0.791	−0.366	0.530	0.780
S_{NH4}	g N m^{-3}	MLP	0.427	0.300	−0.128	0.936	0.968
S_{NH4}	g N m^{-3}	TabNet	0.287	0.248	−0.039	0.971	0.978
S_{NO2}	g N m^{-3}	XGBoost	1.186	1.136	−0.050	0.422	0.470
S_{NO2}	g N m^{-3}	LightGBM	1.203	1.163	−0.040	0.405	0.444
S_{NO2}	g N m^{-3}	CatBoost	1.180	1.134	−0.047	0.428	0.472
S_{NO2}	g N m^{-3}	AdaBoost	1.438	1.407	−0.031	0.151	0.187
S_{NO2}	g N m^{-3}	Random Forest	1.538	1.498	−0.040	0.029	0.078
S_{NO2}	g N m^{-3}	Extra Trees	1.285	1.203	−0.083	0.322	0.406
S_{NO2}	g N m^{-3}	SVR	1.378	1.106	−0.272	0.220	0.498
S_{NO2}	g N m^{-3}	k-NN	1.326	1.274	−0.052	0.278	0.334
S_{NO2}	g N m^{-3}	PLS	1.425	1.377	−0.047	0.167	0.221
S_{NO2}	g N m^{-3}	Multi-task Elastic Net	1.418	1.404	−0.014	0.175	0.190
S_{NO2}	g N m^{-3}	Multi-task Lasso	1.418	1.404	−0.014	0.175	0.190
S_{NO2}	g N m^{-3}	MLP	0.520	0.518	−0.002	0.889	0.890
S_{NO2}	g N m^{-3}	TabNet	0.826	0.823	−0.003	0.720	0.722
S_{NO3}	g N m^{-3}	XGBoost	0.792	0.725	−0.067	0.751	0.792
S_{NO3}	g N m^{-3}	LightGBM	0.789	0.733	−0.057	0.753	0.787
S_{NO3}	g N m^{-3}	CatBoost	0.770	0.708	−0.062	0.765	0.801
S_{NO3}	g N m^{-3}	AdaBoost	1.076	0.983	−0.093	0.541	0.617
S_{NO3}	g N m^{-3}	Random Forest	1.400	1.113	−0.287	0.223	0.509
S_{NO3}	g N m^{-3}	Extra Trees	1.009	0.875	−0.134	0.596	0.696
S_{NO3}	g N m^{-3}	SVR	0.859	0.719	−0.140	0.707	0.795
S_{NO3}	g N m^{-3}	k-NN	1.027	0.768	−0.259	0.582	0.766
S_{NO3}	g N m^{-3}	PLS	1.163	0.747	−0.416	0.464	0.779
S_{NO3}	g N m^{-3}	Multi-task Elastic Net	1.113	0.839	−0.274	0.509	0.721
S_{NO3}	g N m^{-3}	Multi-task Lasso	1.113	0.839	−0.274	0.509	0.721
S_{NO3}	g N m^{-3}	MLP	0.435	0.298	−0.138	0.925	0.965
S_{NO3}	g N m^{-3}	TabNet	0.509	0.294	−0.215	0.897	0.966
S_{N2}	g N m^{-3}	XGBoost	0.790	0.812	+0.022	0.591	0.568
S_{N2}	g N m^{-3}	LightGBM	0.809	0.845	+0.036	0.572	0.533
S_{N2}	g N m^{-3}	CatBoost	0.773	0.807	+0.035	0.609	0.574
S_{N2}	g N m^{-3}	AdaBoost	0.979	0.965	−0.013	0.373	0.390
S_{N2}	g N m^{-3}	Random Forest	1.087	1.031	−0.055	0.227	0.304
S_{N2}	g N m^{-3}	Extra Trees	0.906	0.934	+0.028	0.463	0.429
S_{N2}	g N m^{-3}	SVR	0.990	0.862	−0.128	0.359	0.514
S_{N2}	g N m^{-3}	k-NN	0.872	0.836	−0.036	0.503	0.543
S_{N2}	g N m^{-3}	PLS	1.046	0.974	−0.072	0.285	0.379
S_{N2}	g N m^{-3}	Multi-task Elastic Net	0.995	0.875	−0.120	0.353	0.499
S_{N2}	g N m^{-3}	Multi-task Lasso	0.995	0.875	−0.120	0.353	0.499
S_{N2}	g N m^{-3}	MLP	0.354	0.358	+0.004	0.918	0.916
S_{N2}	g N m^{-3}	TabNet	0.451	0.429	−0.021	0.867	0.879
S_{PO4}	g P m^{-3}	XGBoost	0.352	0.319	−0.033	0.872	0.895
S_{PO4}	g P m^{-3}	LightGBM	0.350	0.321	−0.029	0.874	0.894
S_{PO4}	g P m^{-3}	CatBoost	0.326	0.310	−0.016	0.891	0.901
S_{PO4}	g P m^{-3}	AdaBoost	0.627	0.555	−0.072	0.595	0.683
S_{PO4}	g P m^{-3}	Random Forest	0.840	0.566	−0.274	0.274	0.670
S_{PO4}	g P m^{-3}	Extra Trees	0.691	0.468	−0.222	0.509	0.774
S_{PO4}	g P m^{-3}	SVR	0.471	0.449	−0.022	0.771	0.793

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Table S8 continued

Component	Unit	Surrogate	Raw nRMSE$_j$	Projected nRMSE$_j$	ΔnRMSE$_j$	Raw R_j^2	Projected R_j^2
S_{PO4}	g P m^{-3}	k-NN	0.601	0.498	−0.103	0.628	0.744
S_{PO4}	g P m^{-3}	PLS	0.878	0.684	−0.194	0.206	0.518
S_{PO4}	g P m^{-3}	Multi-task Elastic Net	0.790	0.653	−0.138	0.356	0.561
S_{PO4}	g P m^{-3}	Multi-task Lasso	0.791	0.653	−0.138	0.356	0.561
S_{PO4}	g P m^{-3}	MLP	0.270	0.252	−0.018	0.925	0.935
S_{PO4}	g P m^{-3}	TabNet	0.400	0.352	−0.048	0.835	0.873
S_I	g COD m^{-3}	XGBoost	0.016	0.000	−0.016	1.000	1.000
S_I	g COD m^{-3}	LightGBM	0.004	0.000	−0.004	1.000	1.000
S_I	g COD m^{-3}	CatBoost	0.006	0.000	−0.006	1.000	1.000
S_I	g COD m^{-3}	AdaBoost	0.088	0.000	−0.088	0.992	1.000
S_I	g COD m^{-3}	Random Forest	0.434	0.000	−0.434	0.812	1.000
S_I	g COD m^{-3}	Extra Trees	0.480	0.000	−0.480	0.770	1.000
S_I	g COD m^{-3}	SVR	0.239	0.000	−0.239	0.943	1.000
S_I	g COD m^{-3}	k-NN	0.505	0.000	−0.505	0.745	1.000
S_I	g COD m^{-3}	PLS	0.169	0.000	−0.169	0.971	1.000
S_I	g COD m^{-3}	Multi-task Elastic Net	0.002	0.000	−0.002	1.000	1.000
S_I	g COD m^{-3}	Multi-task Lasso	0.002	0.000	−0.002	1.000	1.000
S_I	g COD m^{-3}	MLP	0.141	0.000	−0.141	0.980	1.000
S_I	g COD m^{-3}	TabNet	0.224	0.000	−0.224	0.950	1.000
S_{ALK}	mol m^{-3}	XGBoost	0.029	0.016	−0.014	0.999	1.000
S_{ALK}	mol m^{-3}	LightGBM	0.027	0.016	−0.011	0.999	1.000
S_{ALK}	mol m^{-3}	CatBoost	0.025	0.015	−0.010	0.999	1.000
S_{ALK}	mol m^{-3}	AdaBoost	0.092	0.026	−0.066	0.992	0.999
S_{ALK}	mol m^{-3}	Random Forest	0.197	0.030	−0.167	0.961	0.999
S_{ALK}	mol m^{-3}	Extra Trees	0.492	0.021	−0.472	0.760	1.000
S_{ALK}	mol m^{-3}	SVR	0.243	0.020	−0.224	0.941	1.000
S_{ALK}	mol m^{-3}	k-NN	0.520	0.023	−0.497	0.732	0.999
S_{ALK}	mol m^{-3}	PLS	0.508	0.024	−0.484	0.745	0.999
S_{ALK}	mol m^{-3}	Multi-task Elastic Net	0.040	0.026	−0.014	0.998	0.999
S_{ALK}	mol m^{-3}	Multi-task Lasso	0.040	0.026	−0.014	0.998	0.999
S_{ALK}	mol m^{-3}	MLP	0.135	0.010	−0.125	0.982	1.000
S_{ALK}	mol m^{-3}	TabNet	0.201	0.007	−0.194	0.960	1.000
X_I	g COD m^{-3}	XGBoost	0.121	0.131	+0.011	0.986	0.983
X_I	g COD m^{-3}	LightGBM	0.116	0.123	+0.007	0.987	0.985
X_I	g COD m^{-3}	CatBoost	0.111	0.119	+0.008	0.988	0.986
X_I	g COD m^{-3}	AdaBoost	0.248	0.286	+0.038	0.939	0.920
X_I	g COD m^{-3}	Random Forest	0.386	0.410	+0.024	0.854	0.835
X_I	g COD m^{-3}	Extra Trees	0.516	0.510	−0.006	0.737	0.743
X_I	g COD m^{-3}	SVR	0.250	0.265	+0.015	0.939	0.931
X_I	g COD m^{-3}	k-NN	0.504	0.505	+0.001	0.750	0.749
X_I	g COD m^{-3}	PLS	0.611	0.629	+0.018	0.632	0.610
X_I	g COD m^{-3}	Multi-task Elastic Net	0.070	0.076	+0.006	0.995	0.994
X_I	g COD m^{-3}	Multi-task Lasso	0.070	0.076	+0.006	0.995	0.994
X_I	g COD m^{-3}	MLP	0.129	0.130	+0.002	0.984	0.983
X_I	g COD m^{-3}	TabNet	0.351	0.357	+0.006	0.879	0.875
X_S	g COD m^{-3}	XGBoost	0.955	0.955	+0.000	0.745	0.745
X_S	g COD m^{-3}	LightGBM	0.920	0.920	+0.000	0.764	0.764
X_S	g COD m^{-3}	CatBoost	0.926	0.926	+0.000	0.760	0.760
X_S	g COD m^{-3}	AdaBoost	1.172	1.187	+0.015	0.616	0.606
X_S	g COD m^{-3}	Random Forest	1.180	1.240	+0.060	0.611	0.570
X_S	g COD m^{-3}	Extra Trees	1.284	1.318	+0.034	0.539	0.514
X_S	g COD m^{-3}	SVR	0.760	0.748	−0.012	0.839	0.844
X_S	g COD m^{-3}	k-NN	1.440	1.423	−0.017	0.421	0.434
X_S	g COD m^{-3}	PLS	1.681	1.456	−0.225	0.210	0.408
X_S	g COD m^{-3}	Multi-task Elastic Net	1.421	1.101	−0.319	0.436	0.661
X_S	g COD m^{-3}	Multi-task Lasso	1.421	1.101	−0.320	0.436	0.661
X_S	g COD m^{-3}	MLP	0.342	0.342	−0.000	0.967	0.967
X_S	g COD m^{-3}	TabNet	0.344	0.340	−0.004	0.967	0.968
X_H	g COD m^{-3}	XGBoost	0.460	0.786	+0.326	0.859	0.589
X_H	g COD m^{-3}	LightGBM	0.460	0.749	+0.290	0.860	0.627
X_H	g COD m^{-3}	CatBoost	0.447	0.719	+0.272	0.867	0.656
X_H	g COD m^{-3}	AdaBoost	0.861	1.386	+0.525	0.507	−0.276
X_H	g COD m^{-3}	Random Forest	0.688	2.653	+1.965	0.686	−3.677
X_H	g COD m^{-3}	Extra Trees	0.966	1.519	+0.552	0.379	−0.533
X_H	g COD m^{-3}	SVR	0.502	0.737	+0.235	0.832	0.639
X_H	g COD m^{-3}	k-NN	0.741	1.319	+0.578	0.635	−0.157
X_H	g COD m^{-3}	PLS	1.159	0.996	−0.163	0.108	0.341
X_H	g COD m^{-3}	Multi-task Elastic Net	0.912	0.774	−0.138	0.447	0.602
X_H	g COD m^{-3}	Multi-task Lasso	0.912	0.774	−0.139	0.447	0.602
X_H	g COD m^{-3}	MLP	0.258	0.375	+0.117	0.956	0.907
X_H	g COD m^{-3}	TabNet	0.311	0.569	+0.259	0.936	0.784
X_{PAO}	g COD m^{-3}	XGBoost	0.196	0.209	+0.013	0.959	0.953
X_{PAO}	g COD m^{-3}	LightGBM	0.205	0.216	+0.011	0.955	0.950
X_{PAO}	g COD m^{-3}	CatBoost	0.192	0.208	+0.016	0.961	0.954
X_{PAO}	g COD m^{-3}	AdaBoost	0.291	0.295	+0.004	0.910	0.907
X_{PAO}	g COD m^{-3}	Random Forest	0.600	0.608	+0.008	0.615	0.605
X_{PAO}	g COD m^{-3}	Extra Trees	0.415	0.419	+0.005	0.816	0.812
X_{PAO}	g COD m^{-3}	SVR	0.312	0.312	−0.000	0.896	0.896
X_{PAO}	g COD m^{-3}	k-NN	0.507	0.498	−0.009	0.726	0.735
X_{PAO}	g COD m^{-3}	PLS	0.706	0.741	+0.035	0.467	0.413

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Table S8 continued

Component	Unit	Surrogate	Raw nRMSE$_j$	Projected nRMSE$_j$	ΔnRMSE$_j$	Raw R_j^2	Projected R_j^2
X_{PAO}	g COD m^{-3}	Multi-task Elastic Net	0.318	0.288	−0.030	0.892	0.912
X_{PAO}	g COD m^{-3}	Multi-task Lasso	0.318	0.288	−0.030	0.892	0.912
X_{PAO}	g COD m^{-3}	MLP	0.197	0.198	+0.001	0.959	0.958
X_{PAO}	g COD m^{-3}	TabNet	0.285	0.289	+0.004	0.913	0.911
X_{PP}	g P m^{-3}	XGBoost	0.391	0.407	+0.015	0.824	0.811
X_{PP}	g P m^{-3}	LightGBM	0.391	0.408	+0.016	0.824	0.810
X_{PP}	g P m^{-3}	CatBoost	0.380	0.388	+0.008	0.834	0.827
X_{PP}	g P m^{-3}	AdaBoost	0.657	0.715	+0.058	0.505	0.413
X_{PP}	g P m^{-3}	Random Forest	0.710	0.783	+0.073	0.421	0.297
X_{PP}	g P m^{-3}	Extra Trees	0.605	0.620	+0.015	0.580	0.559
X_{PP}	g P m^{-3}	SVR	0.549	0.545	−0.004	0.654	0.659
X_{PP}	g P m^{-3}	k-NN	0.588	0.619	+0.031	0.603	0.561
X_{PP}	g P m^{-3}	PLS	0.968	0.872	−0.097	−0.075	0.129
X_{PP}	g P m^{-3}	Multi-task Elastic Net	0.876	0.813	−0.063	0.120	0.243
X_{PP}	g P m^{-3}	Multi-task Lasso	0.876	0.813	−0.064	0.120	0.243
X_{PP}	g P m^{-3}	MLP	0.315	0.312	−0.003	0.886	0.888
X_{PP}	g P m^{-3}	TabNet	0.454	0.429	−0.025	0.764	0.789
X_{PHA}	g COD m^{-3}	XGBoost	0.469	0.468	−0.001	0.785	0.785
X_{PHA}	g COD m^{-3}	LightGBM	0.477	0.476	−0.001	0.777	0.778
X_{PHA}	g COD m^{-3}	CatBoost	0.472	0.471	−0.002	0.781	0.783
X_{PHA}	g COD m^{-3}	AdaBoost	0.786	0.787	+0.001	0.395	0.393
X_{PHA}	g COD m^{-3}	Random Forest	0.793	0.793	+0.000	0.384	0.383
X_{PHA}	g COD m^{-3}	Extra Trees	0.682	0.682	+0.000	0.545	0.545
X_{PHA}	g COD m^{-3}	SVR	0.710	0.705	−0.005	0.507	0.513
X_{PHA}	g COD m^{-3}	k-NN	0.698	0.698	−0.000	0.522	0.523
X_{PHA}	g COD m^{-3}	PLS	1.057	0.813	−0.244	−0.095	0.352
X_{PHA}	g COD m^{-3}	Multi-task Elastic Net	1.031	0.719	−0.313	−0.042	0.494
X_{PHA}	g COD m^{-3}	Multi-task Lasso	1.032	0.719	−0.313	−0.043	0.494
X_{PHA}	g COD m^{-3}	MLP	0.350	0.328	−0.022	0.880	0.895
X_{PHA}	g COD m^{-3}	TabNet	0.550	0.499	−0.052	0.703	0.756
X_{AOB}	g COD m^{-3}	XGBoost	0.427	0.418	−0.009	0.880	0.885
X_{AOB}	g COD m^{-3}	LightGBM	0.430	0.421	−0.008	0.878	0.883
X_{AOB}	g COD m^{-3}	CatBoost	0.411	0.402	−0.009	0.889	0.894
X_{AOB}	g COD m^{-3}	AdaBoost	0.804	0.796	−0.008	0.574	0.582
X_{AOB}	g COD m^{-3}	Random Forest	1.158	1.133	−0.025	0.115	0.153
X_{AOB}	g COD m^{-3}	Extra Trees	0.746	0.731	−0.014	0.633	0.647
X_{AOB}	g COD m^{-3}	SVR	0.552	0.549	−0.003	0.799	0.801
X_{AOB}	g COD m^{-3}	k-NN	0.719	0.706	−0.013	0.659	0.671
X_{AOB}	g COD m^{-3}	PLS	0.910	0.908	−0.001	0.455	0.456
X_{AOB}	g COD m^{-3}	Multi-task Elastic Net	0.725	0.716	−0.010	0.653	0.662
X_{AOB}	g COD m^{-3}	Multi-task Lasso	0.726	0.716	−0.010	0.653	0.662
X_{AOB}	g COD m^{-3}	MLP	0.245	0.244	−0.001	0.961	0.961
X_{AOB}	g COD m^{-3}	TabNet	0.364	0.361	−0.003	0.912	0.914
X_{NOB}	g COD m^{-3}	XGBoost	0.291	0.289	−0.003	0.920	0.922
X_{NOB}	g COD m^{-3}	LightGBM	0.296	0.293	−0.003	0.918	0.919
X_{NOB}	g COD m^{-3}	CatBoost	0.281	0.279	−0.002	0.925	0.926
X_{NOB}	g COD m^{-3}	AdaBoost	0.481	0.475	−0.006	0.782	0.787
X_{NOB}	g COD m^{-3}	Random Forest	0.948	0.936	−0.012	0.153	0.174
X_{NOB}	g COD m^{-3}	Extra Trees	0.546	0.540	−0.006	0.719	0.725
X_{NOB}	g COD m^{-3}	SVR	0.468	0.466	−0.002	0.794	0.796
X_{NOB}	g COD m^{-3}	k-NN	0.549	0.545	−0.005	0.716	0.721
X_{NOB}	g COD m^{-3}	PLS	0.942	0.933	−0.010	0.164	0.181
X_{NOB}	g COD m^{-3}	Multi-task Elastic Net	0.499	0.490	−0.008	0.766	0.774
X_{NOB}	g COD m^{-3}	Multi-task Lasso	0.499	0.490	−0.008	0.766	0.774
X_{NOB}	g COD m^{-3}	MLP	0.225	0.225	−0.001	0.952	0.952
X_{NOB}	g COD m^{-3}	TabNet	0.308	0.308	−0.000	0.910	0.911
X_{MeP}	g TSS m^{-3}	XGBoost	0.255	0.241	−0.013	0.936	0.942
X_{MeP}	g TSS m^{-3}	LightGBM	0.239	0.233	−0.006	0.943	0.946
X_{MeP}	g TSS m^{-3}	CatBoost	0.230	0.223	−0.007	0.948	0.951
X_{MeP}	g TSS m^{-3}	AdaBoost	0.573	0.450	−0.123	0.674	0.799
X_{MeP}	g TSS m^{-3}	Random Forest	0.941	0.555	−0.386	0.121	0.695
X_{MeP}	g TSS m^{-3}	Extra Trees	0.788	0.420	−0.368	0.384	0.825
X_{MeP}	g TSS m^{-3}	SVR	0.395	0.321	−0.074	0.845	0.898
X_{MeP}	g TSS m^{-3}	k-NN	0.565	0.385	−0.180	0.684	0.853
X_{MeP}	g TSS m^{-3}	PLS	0.890	0.453	−0.437	0.215	0.797
X_{MeP}	g TSS m^{-3}	Multi-task Elastic Net	0.468	0.451	−0.016	0.783	0.798
X_{MeP}	g TSS m^{-3}	Multi-task Lasso	0.468	0.451	−0.016	0.783	0.798
X_{MeP}	g TSS m^{-3}	MLP	0.249	0.223	−0.025	0.939	0.950
X_{MeP}	g TSS m^{-3}	TabNet	0.393	0.288	−0.106	0.846	0.918
X_{MeOH}	g TSS m^{-3}	XGBoost	0.427	0.404	−0.023	0.861	0.876
X_{MeOH}	g TSS m^{-3}	LightGBM	0.410	0.390	−0.020	0.872	0.884
X_{MeOH}	g TSS m^{-3}	CatBoost	0.383	0.373	−0.010	0.888	0.894
X_{MeOH}	g TSS m^{-3}	AdaBoost	0.771	0.753	−0.018	0.547	0.568
X_{MeOH}	g TSS m^{-3}	Random Forest	1.048	0.929	−0.120	0.162	0.343
X_{MeOH}	g TSS m^{-3}	Extra Trees	0.826	0.703	−0.123	0.479	0.623
X_{MeOH}	g TSS m^{-3}	SVR	0.579	0.537	−0.042	0.744	0.780
X_{MeOH}	g TSS m^{-3}	k-NN	0.727	0.644	−0.083	0.597	0.684
X_{MeOH}	g TSS m^{-3}	PLS	0.971	0.758	−0.213	0.281	0.562
X_{MeOH}	g TSS m^{-3}	Multi-task Elastic Net	0.783	0.755	−0.028	0.532	0.565
X_{MeOH}	g TSS m^{-3}	Multi-task Lasso	0.783	0.755	−0.028	0.532	0.565

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Table S8 continued

Component	Unit	Surrogate	Raw nRMSE$_j$	Projected nRMSE$_j$	ΔnRMSE$_j$	Raw R_j^2	Projected R_j^2
X_{MeOH}	g TSS m^{-3}	MLP	0.384	0.374	-0.010	0.888	0.893
X_{MeOH}	g TSS m^{-3}	TabNet	0.543	0.481	-0.062	0.775	0.823

Table S9: Severe out-of-distribution accuracy by derived water-quality composite and surrogate. Composite nRMSE is physical-unit RMSE divided by the population SD of that composite in all 10,000 in-distribution targets. Each value summarizes the same 600 mechanistic cases; Δ is projected minus raw.

Composite	Surrogate	Raw nRMSE$_q$	Projected nRMSE$_q$	ΔnRMSE$_q$	Raw R^2	Projected R^2
COD	XGBoost	0.671	0.783	+0.112	0.786	0.709
COD	LightGBM	0.654	0.752	+0.098	0.796	0.731
COD	CatBoost	0.654	0.742	+0.088	0.797	0.738
COD	AdaBoost	0.863	1.137	+0.274	0.646	0.385
COD	Random Forest	0.915	1.836	+0.921	0.602	−0.603
COD	Extra Trees	1.083	1.283	+0.200	0.442	0.217
COD	SVR	0.431	0.604	+0.173	0.911	0.826
COD	k-NN	0.952	1.155	+0.203	0.569	0.366
COD	PLS	1.047	0.945	−0.102	0.478	0.575
COD	Multi-task Elastic Net	0.448	0.644	+0.196	0.904	0.803
COD	Multi-task Lasso	0.448	0.644	+0.196	0.904	0.803
COD	MLP	0.219	0.287	+0.068	0.977	0.961
COD	TabNet	0.282	0.373	+0.091	0.962	0.934
TN	XGBoost	1.043	0.394	−0.649	0.620	0.946
TN	LightGBM	1.029	0.410	−0.618	0.631	0.941
TN	CatBoost	1.038	0.392	−0.646	0.624	0.946
TN	AdaBoost	1.136	0.469	−0.668	0.549	0.923
TN	Random Forest	1.956	0.501	−1.455	−0.336	0.912
TN	Extra Trees	1.503	0.454	−1.050	0.211	0.928
TN	SVR	0.798	0.419	−0.379	0.778	0.939
TN	k-NN	1.319	0.406	−0.913	0.393	0.942
TN	PLS	1.523	0.473	−1.050	0.190	0.922
TN	Multi-task Elastic Net	0.484	0.425	−0.059	0.918	0.937
TN	Multi-task Lasso	0.484	0.425	−0.059	0.918	0.937
TN	MLP	0.426	0.174	−0.253	0.937	0.989
TN	TabNet	0.558	0.209	−0.349	0.891	0.985
TP	XGBoost	0.144	0.000	−0.143	0.978	1.000
TP	LightGBM	0.119	0.000	−0.119	0.985	1.000
TP	CatBoost	0.105	0.000	−0.105	0.988	1.000
TP	AdaBoost	0.383	0.000	−0.382	0.847	1.000
TP	Random Forest	0.936	0.000	−0.936	0.081	1.000
TP	Extra Trees	0.732	0.000	−0.732	0.438	1.000
TP	SVR	0.231	0.000	−0.231	0.944	1.000
TP	k-NN	0.497	0.000	−0.497	0.740	1.000
TP	PLS	0.732	0.000	−0.732	0.438	1.000
TP	Multi-task Elastic Net	0.003	0.000	−0.003	1.000	1.000
TP	Multi-task Lasso	0.003	0.000	−0.003	1.000	1.000
TP	MLP	0.126	0.000	−0.126	0.983	1.000
TP	TabNet	0.294	0.000	−0.294	0.910	1.000
TSS	XGBoost	0.678	0.826	+0.148	0.768	0.656
TSS	LightGBM	0.660	0.788	+0.128	0.780	0.687
TSS	CatBoost	0.659	0.779	+0.121	0.781	0.694
TSS	AdaBoost	0.878	1.232	+0.354	0.611	0.234
TSS	Random Forest	0.955	2.119	+1.164	0.540	−1.264
TSS	Extra Trees	1.152	1.420	+0.267	0.330	−0.017
TSS	SVR	0.477	0.661	+0.184	0.885	0.780
TSS	k-NN	0.922	1.221	+0.299	0.571	0.248
TSS	PLS	1.199	1.057	−0.142	0.275	0.436
TSS	Multi-task Elastic Net	0.412	0.690	+0.277	0.914	0.760
TSS	Multi-task Lasso	0.413	0.690	+0.277	0.914	0.760
TSS	MLP	0.275	0.333	+0.058	0.962	0.944
TSS	TabNet	0.379	0.522	+0.144	0.928	0.862

Every raw out-of-distribution prediction set violated mass conservation. Projection eliminated mass-conservation and non-negativity violations across all 15,600 out-of-distribution surrogate predictions, with maximum projected conservation residual 1.72×10^{-13}. Positivity backtracking was activated 29 times under mild and 39 times under severe extrapolation.

7 Full-Data Hyperparameters

Table S10 records the selected settings used for the final full in-distribution refits and out-of-distribution evaluation. These settings were selected in separate four-fold, 100-trial studies using only in-distribution data.

Table S10: Selected settings for the final full-data surrogate refits used in out-of-distribution evaluation. Fixed implementation controls are included because they form part of the specified fitting protocol.

Surrogate	Selected full-data settings
XGBoost	colsample__bytree=0.9560581387610049, device="cpu", learning__rate=0.061689489299135404, max__depth=7, min__child__weight=2.90921863007842, n__estimators=247, n__jobs=1, objective="reg:squarederror", random__state=42, reg__alpha=0.36819809305976897, reg__lambda=7.203558426205165, subsample=0.7391736712339139, tree__method="hist", verbosity=0

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Table S10 continued

Surrogate	Selected full-data settings
LightGBM	colsample_bytree=0.9842299479772286, device_type="cpu", learning_rate=0.041557555636326, max_depth=10, min_child_samples=8, n_estimators=231, n_jobs=1, num_leaves=56, objective="regression", random_state=42, reg_alpha=0.00017273364472452597, reg_lambda=0.5483223255853542, subsample=0.663454486337347, subsample_freq=1, verbosity=-1
CatBoost	allow_writing_files=false, bagging_temperature=0.349485498928751, bootstrap_type="Bayesian", depth=8, iterations=304, l2_leaf_reg=0.6057379580810124, learning_rate=0.10301323327550004, loss_function="RMSE", random_seed=42, random_strength=0.0010555159674713857, task_type="CPU", thread_count=1, verbose=false
AdaBoost	learning_rate=0.11207319434076271, loss="exponential", n_estimators=110, random_state=42
Random Forest	bootstrap=false, max_depth=null, max_features=0.5283540452009149, min_samples_leaf=1, min_samples_split=2, n_estimators=400, n_jobs=12, random_state=42
Extra Trees	bootstrap=false, max_depth=18, max_features=0.9699598005443838, min_samples_leaf=1, min_samples_split=2, n_estimators=351, n_jobs=12, random_state=42
SVR	C=6.287447897273593, cache_size=256, epsilon=0.010017341146432681, gamma="scale", kernel="rbf", max_iter=20000, shrinking=true, tol=0.007355017843876341
k-NN	algorithm="kd_tree", leaf_size=44, metric="euclidean", n_jobs=12, n_neighbors=15, p=1, weights="distance"
PLS	copy=true, max_iter=735, n_components=6, scale=false, tol=1.0806007264903645e-08
Multi-task Elastic Net	alpha=0.0023048512732013536, fit_intercept=true, l1_ratio=0.9443842016406682, max_iter=20000, random_state=42, selection="cyclic", tol=1e-06
Multi-task Lasso	alpha=0.0020826635913711827, fit_intercept=true, max_iter=20000, random_state=42, selection="cyclic", tol=1e-06
MLP	activation="tanh", alpha=3.256832487981403e-05, batch_size=128, early_stopping=false, hidden_layer_sizes=[128, 64, 32], learning_rate_init=0.001761406372323539, max_iter=250, n_iter_no_change=15, random_state=42, shuffle=true, solver="adam", tol=0.00010997527135720564, verbose=false
TabNet	batch_size=1024, device_name="cpu", gamma=1.179749350809977, lambda_sparse=3.365610485088621e-05, learning_rate=0.020324962775784057, mask_type="entmax", max_epochs=49, n_steps=3, seed=42, verbose=0, virtual_batch_size=128, width=24